Environmental Sciences Europe

REVIEW

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Harnessing bacterial consortia for effective bioremediation: targeted removal of heavy metals, hydrocarbons, and persistent pollutants

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Abstract

Industrial activities, agricultural runoff, municipal waste, and mining generate toxic pollutants that threaten ecosystems and human health, necessitating sustainable remediation strategies to mitigate their impact. Bacterial bioremediation is an eco-friendly and cost-effective method for treating metal-contaminated industrial effluent. It uses biosorption and bioaccumulation mechanisms, redox reactions, and enzymatic transformation methods, with bacterial cell walls as potential chemisorption sites. Simultaneously, bacteria employing various ways to tolerate and detoxify metalloid pollutants play a crucial role in mitigating hazards. This review highlights the most effective bacterial consortia for removing heavy metals, hydrocarbons, and persistent pollutants, emphasizing their mechanisms and applications. In addition, it explores emerging technologies, including synthetic biology, genetic engineering, nanotechnology, and CRISPR-Cas9, to enhance biodegradation efficiency, particularly in underexplored areas like plastic decomposition. These advancements hold significant promises for improving bioremediation efficacy and expanding its industrial and environmental applications.

Keywords Microbial bioremediation, Bacterial consortia, Heavy metal removal, Hydrocarbon degradation, Synthetic biology, CRISPR-engineered microbes, Nanobioremediation

Introduction

Microbes exhibit a remarkable variety of metabolic pathways, enabling them to decompose several pollutants. Bacteria that break down hydrocarbons, such as Pseudomonas and *Bacillus spe*cies, make enzymes such as hydroxylases and dioxygenases that break down petroleum compounds [26, 86, 87]. These enzymes catalyze the transformation of hydrocarbons into less harmful byproducts, such as carbon dioxide and water. In addition, white rot fungi use ligninolytic enzymes to break down persistent pollutants such as PCBs and pesticides [189, 190].

Biological enhancement and biostimulation can enhance the bioremediation that microbes facilitate. The process of bioaugmentation involves adding specific microorganisms to polluted areas to make the existing microbial community work better [70]. The incorporation of *Pseudomonas putida* into oil-contaminated soil can expedite hydrocarbon breakdown [407]. On the other hand, biostimulation focuses on improving the environment to bring back to life native microbial populations. This is usually done by changing pH levels, adding important nutrients, or keeping oxygen levels at the right level [174]. Syntrophic, an essential process in microbial bioremediation, entails the collaboration of many microbial species to decompose intricate pollutants [142]. Syntrophic bacteria collaborate to decompose substances such as chlorinated solvents or aromatic hydrocarbons in anaerobic settings. Co-metabolism transpires when a microbe metabolizes a molecule as a secondary

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reaction while using another primary substrate. This effect expands the range of pollutants susceptible to remediation, as bacteria inadvertently break down toxins as part of their metabolic processes [230].

Recent advances in molecular biology and genetic engineering have enabled the alteration of strains of bacteria to improve bioremediation efficacy (Zhou et al., 2020). Genetic methodologies such as recombinant DNA technology enable scientists to modify bacteria with certain enzymes or pathways for the precise breakdown of contaminants [220]. Genetically modified microorganisms (GMOs) have shown promise in breaking down persistent organic pollutants, such as chlorinated substances and endocrine-disrupting toxins [368–370, 372].

Bioremediation is a beneficial method that removes the toxic effects of metalloids and recovers them for industrial applications. It outperforms other methods in eradicating metalloid contamination in various environments [195]. Microbial bioremediation is gaining interest due to its cost-effectiveness, eco-friendliness, and simplicity. Bacteria play a crucial role due to their small size, rapid growth, and adaptability. Factors affecting bioremediation include soil type, contaminants, nutrient availability, pH, and temperature [192, 401, 402]. Mycoremediation employs fungi for the removal of metalloids, whereas phytoremediation uses algae and microalgae. Phytoremediation of contaminated areas employs endophytic microorganisms [177].

Human activity, including industrialization, contributes to environmental pollution through inhalation, oral absorption, and ingestion, causing health effects and varying doses depending on exposure duration and intensity [225]. In addition to industrial effluents, contamination arises from agricultural pesticides, mining tailings, municipal solid waste leachates, and atmospheric deposition, contributing to the widespread presence of heavy metals, hydrocarbons, and persistent pollutants in ecosystems [120]. A comprehensive understanding of pollution sources is critical for designing targeted bioremediation strategies. Apart from industrial wastewater, major contributors include agrochemical residues, landfill leachates, and urban runoff, all of which introduce persistent pollutants into ecosystems [315].

Environmental pollution, primarily from automotive emissions, power generation facilities, waste incineration, chemical manufacturing firms, and volcanic eruptions, releases hazardous substances such as SO_2, CO_2, NO_2, heavy metals, biological pollutants, ozone, and tobacco smoke into the atmosphere, causing health risks such as cancer, cardiovascular, reproductive, and respiratory issues [329]. Tobacco smoke, containing harmful chemicals, can cause health issues such as cancer, asthma, bronchitis, pharyngitis, and ocular irritation. Exposure to biological contaminants, including bacteria, viruses, and pollen, can cause allergic diseases and irritation. Prolonged exposure can damage the liver, brain, and digestive systems and potentially lead to cancer [68]. Exposure to ozone can result in ocular irritation, burns, and respiratory ailments. Prolonged exposure may result in neurological, reproductive, respiratory, and oncological complications. Winter exposure can also cause respiratory problems [123]. Persistent organic pollutants (POPs) are toxic chemicals causing global health and environmental harm. Wind and water can transport them, they can persist for long periods, and they can pass through the food chain [259].

Unlike previous reviews, this study uniquely emphasizes the role of bacterial consortia in bioremediation, integrating emerging trends such as CRISPR-based bacterial engineering, synthetic microbial consortia, and nanobioremediation to offer a comprehensive perspective on next-generation bioremediation strategies. While earlier studies have explored microbial bioremediation in general, this review focuses on how engineered bacterial systems and novel biotechnological interventions can improve pollutant degradation efficiency. Moreover, it highlights underexplored areas such as plastic degradation and bioremediation in extreme environments, providing a forward-looking perspective on the field. This review explores the use of specific bacteria for targeted bioremediation of pollutants, examining both established and emerging techniques, and emphasizing synthetic biology innovations that could revolutionize environmental cleanup efforts.

Mechanisms of bacterial bioremediation

Bioremediation is a biological process that removes hazardous pollutants from the environment without generating secondary contaminants [314]. Bacteria are particularly effective in bioremediating metal ions due to their adaptability and versatility [309]. These microorganisms have developed specialized mechanisms to survive in toxic environments, including unique membrane properties, the production of extra polymeric substances (EPS), siderophores, and enzyme/protein-based protective systems [136, 173].

Several bacterial species have been extensively studied for their ability to remediate heavy metal ions through different mechanisms. For instance, genetically engineered microorganisms and immobilized microbial cells have been employed for heavy metal removal using processes such as adsorption, biological leaching, biomineralization, biotransformation, and cellular deposition [264]. Some bacteria demonstrate a remarkable capacity for metal ion storage. For example, *Vibrio harveyi*, commonly found in marine environments, can accumulate cadmium ions up to 23.3 mg $Cd^{2}+$/g of dry biomass, making it a promising candidate for cadmium bioremediation [1]. Similarly, mercury ions can be efficiently remediated by bacterial consortia, highlighting the effectiveness of microbial cooperation in detoxification [292].

The ability of bacteria to biosorb, transform, and tolerate heavy metals such as zinc, lead, cadmium, and copper is attributed to metal immobilization and mobilization mechanisms [169]. These mechanisms enable bacterial survival in metal-contaminated environments while facilitating the removal of toxic heavy metals. The efficiency of bioremediation is influenced by factors such as temperature, pH, contaminant concentration, oxygen availability, and agitation [157].

Microorganisms employ two primary approaches for contaminant removal: immobilization and mobilization [96, 362]. Mobilization strategies include enzymatic oxidation, bioleaching, biological stimulation, bioaugmentation, and enzymatic reduction, which facilitate the breakdown and removal of contaminants. In contrast, immobilization strategies, such as metabolism, complexation, absorption, and solidification, render pollutants inaccessible within the environment [24]. During mineralization, microorganisms convert contaminants into final products such as carbon dioxide and water, effectively detoxifying the environment. Meanwhile, immobilization mechanisms, such as nitrate–nitrogen transformation, prevent pollutant dispersion, making them particularly useful for remediating heavily contaminated sites [286, 304].

Immobilization techniques can be executed through in-situ or ex-situ approaches [286]. The ex-situ method involves transporting contaminated soil to a different location, where microbial processes immobilize the metal ions responsible for pollution [23]. Conversely, in the in-situ method, bioremediation occurs directly at the contamination site, making it a more practical approach in certain scenarios [53]. Specific bacterial species, such as *Enterobacter asburiae* and *Bacillus cereus*, have demonstrated efficiency in removing heavy metals from polluted environments, making them valuable candidates for bioremediation applications [114]. In addition, bacteria protect themselves from toxic substances by producing hydrophobic or solvent efflux pumps that form a defensive barrier on the cell's outer membrane [362].

Enzymatic oxidation is another essential bacterial bioremediation mechanism that reduces hazardous heavy metals, dyes, and phenols by altering their oxidation states [387]. This process generates free radicals, which subsequently decompose into various fractions, producing larger molecular-weight molecules [356]. One key enzyme involved in this process is laccase, an oxidoreductase enzyme that catalyzes the oxidation of aromatic amines, phenols, and polyphenols, facilitating the reduction of molecular oxygen to water [117, 312]. Laccase production has been observed in *Pycnoporus* sp. and *Leptosphaerulina* sp., demonstrating their potential in heavy metal digestion [352].

Enzymatic reduction, another bacterial bioremediation strategy, is primarily carried out by obligate and facultative anaerobes, transforming pollutants into a reduced oxidation state that renders them insoluble. This mechanism is particularly effective in remediating polychlorinated dibenzo-p-dioxins and dibenzofurans [25]. In addition, enzymes such as chrome reductase facilitate the reduction of hexavalent chromium to trivalent chromium, while azoreductase breaks down azo compounds by cleaving azo bonds [317].

Heavy metal co-contamination in crude oil-polluted environments can hinder microbial bioremediation of hydrocarbons. However, certain bacterial species have demonstrated resilience by simultaneously resisting heavy metals and degrading hydrocarbons. For example, *Cupriavidus metallidurans* CH34 exhibits resistance to cadmium and mercury while efficiently degrading benzene, making it a promising candidate for bioremediation in metal-contaminated petroleum sites [18]. Similarly, *Bacillus cereus* M6 employs efflux pumps and intracellular accumulation mechanisms to remove Pb(II), Cr(VI), and As(V), demonstrating its potential for detoxifying multi-metal environments [328]. In addition, *Delftia lacustris* LZ-C has shown the ability to degrade hydrocarbons while exhibiting significant resistance to Cr(VI), Hg(II), Cd(II), and Pb(II), making it a strong candidate for remediation in metal-polluted areas [381].

Exopolysaccharide-producing *Bacillus* strains have proven highly effective in adsorbing cadmium and lead from contaminated soils, enhancing plant growth while reducing heavy metal bioavailability [400, 404]. Various microorganisms, including bacteria, fungi, and actinomycetes, are known to synthesize exopolysaccharides, with *Bacillus*, *Pseudomonas*, *Streptococcus*, and *Lactobacillus* being among the key producers. Exopolysaccharides play a crucial role in heavy metal bioremediation by facilitating complexation, ion exchange, precipitation, and redox reactions. For instance, *Halomonas* sp. mitigates arsenic toxicity by oxidizing As^{3+} to the less toxic As^{6+}, thereby aiding in soil detoxification and improving rice growth [400, 404].

Research by Hernández-Guerrero et al. [144] highlighted the efficiency of *Bacillus zhejiangensis* CEIB S4-3 in removing approximately 90% of Cd^{2+} and 91% of Pb^{2+} (50 mg/L) when tested individually. However, in a Cd–Pb mixture, removal rates dropped to 59% for Cd and 75% for Pb, with Cd primarily removed via extracellular biosorption and Pb via intracellular bioaccumulation.

Genome analysis revealed the presence of genes related to heavy metal resistance, including those involved in sensing, transcriptional responses, and efflux systems. In addition, *Bacillus cereus* strains BRS 11 and BRS 17 displayed high tolerance to heavy metals, effectively reducing Cr(VI) and Ni(II) using electron donors and exhibiting significant synergistic resistance [100].

Enterobacter kobei FACU6 has also demonstrated remarkable bioremediation potential, efficiently removing lead (83.4%, 571.9 mg/g) with a maximum tolerable concentration (MTC) of 3000 mg/L. Genome analysis identified key heavy metal resistance genes, which were significantly upregulated under specific environmental conditions, reinforcing its suitability for bioremediation applications [105]. Furthermore, polyextremophilic bacteria such as *Pseudomonas aeruginosa* san ai exhibit exceptional bioremediation efficiency, degrading a wide range of petroleum hydrocarbons, including n-alkanes and polycyclic aromatic hydrocarbons, with up to 98% efficiency. This strain employs terminal oxidation, salicylate, and phthalate pathways to break down contaminants, making it highly effective in extreme environments with variable pH and heavy metal concentrations [233].

Another notable bioremediator is *Pseudoxanthomonas mexicana* GTZY, a wastewater bacterium capable of transforming mercury and arsenic. Abdul Raheem et al. [3] reported that GTZY removed 84.3% of mercury through Hg(II) volatilization, confirmed via *merA* gene amplification. In addition, it achieved 68.5% removal of arsenate and 63.2% of arsenite, with resistance levels exceeding 175 mM and 55 mM, respectively. Genome analysis revealed that the *mer* operon was associated with a Tn2-transposon, suggesting horizontal gene transfer, while the *ars* operon displayed unique diversification. These findings highlight the diverse bacterial mechanisms that optimize bioremediation processes in environments contaminated by heavy metals and hydrocarbons. By understanding these mechanisms, more effective strategies could be developed to clean up polluted ecosystems.

Specific bacteria and their bioremediation potential

Physicochemical techniques are ineffective in removing contaminants under extreme conditions due to the reduced availability of harmful microorganisms (HMs) and secondary contamination threats. Integrating extremophiles in biotechnology is crucial for sustainable bioremediation processes for HMs [71]. Bacterial bioremediation effectively removes HMs from water, a growing concern due to industrialization, using bacteria that are environmentally friendly and tolerant to various HMs [344].

Pseudomonas aeruginosa

Research indicates that *P. aeruginosa* can break down petroleum hydrocarbons, potentially eliminating "up to 95.71% of total petroleum hydrocarbons (TPH) in crude oil" [38, 234, 235]. This shows that *P. aeruginosa* has important possibilities for bioremediation of this organic pollution. They have categorized and presented in Table 1 the *P. aeruginosa* strains identified to date that exhibit petroleum hydrocarbon-degrading capabilities. Petroleum hydrocarbons refer to chemical contaminants originating from petroleum, encompassing both aliphatic and aromatic molecules [297]. Because of how complex petroleum is, *P. aeruginosa* strains that are grown on crude oil as their only source of carbon usually could break down a wide range of petroleum hydrocarbons. Removal studies indicate that short-chain compounds are easier to eliminate than long-chain hydrocarbons and that aliphatic hydrocarbons remove more quickly than aromatic ones [301]. The augmentation of fused ring structures enhances hydrophobicity and diminishes solubility, hence decreasing the accessibility of these substances for microbiological breakdown [234, 235].

Bioavailability is crucial for removal efficacy, and *P. aeruginosa* biosurfactants, such as rhamnolipids and lipopeptides, contribute to petroleum hydrocarbon removal when grown on petroleum hydrocarbons as the sole carbon source [237, 301]. After 12 h of cultivation on crude oil, *P. aeruginosa* SNP0614 produced a lipopeptide that reduced the surface tension to 25.4 mN/m. As shown by the 24-h emulsification index (E24) [208], the lipopeptide was 90% effective at emulsifying crude oil after 20 days of incubation. By emulsifying them, biosurfactants make petroleum hydrocarbons easier to dissolve and spread out, which makes them more bioavailable to *P. aeruginosa* [149].

Bacillus subtilis

Heavy metal pollution from human activities and industrialization threatens ecosystems and humans, affecting water, soil, and air. Even at low concentrations, these reactive elements cause metabolic disturbances, infertility, diminished immunological systems, stunted growth, and potentially fatal disorders [122, 241]. Arsenic, boron, cadmium, copper, mercury, molybdenum, nickel, lead, chromium, and zinc are major polluting metals with toxic effects on plants, animals, and humans. High concentrations cause environmental and health problems [85]. Bioremediation processes can minimize utilizing living organisms and enzymes to bioremove toxic molecules into less harmful substances and can mitigate environmental issues resulting from heavy metals [197].

Bacillus subtilis, a Gram-positive bacterium, is known for its high efficiency in removing heavy metals and its

Table 1 *Pseudomonas aeruginosa* related to the breakdown of petroleum hydrocarbons

Substrate	Substrate composition	Strain	Source	Decomposition duration	Rate of removal	References
Diesel oil	2%	*P. aeruginosa* DSM50071	Soil	20 days	TPH: 87%, C_{10}: 90%, C_{20+}: 69%, AH: 47%	Safdari et al. [310]
	25%	*P. aeruginosa InaCC B290*	–	25 days	TPH: 36.50%, $C_{14}\sim C_{20}$: 37.67–60.36%	Safitri et al. [310]
Crude petroleum-oil hydrocarbon	5%	*P. aeruginosa* (NCIB 950)	–	18 days	TPH: 99%	Adeyi et al. [5]
Petroleum	2%	*P. aeruginosa* AT 18	Soil	15 days	TPH 84%	Perez et al. [279]
Crude oil/motor oil	50,000 ppm; 5%	*P. aeruginosa* PP3 and *P. aeruginosa* PP4	–	40 days	TPH 52% (CO), 67% (MO)	Muthukumar et al. [246]
Crude and mustard oils	300 µL	*P. aeruginosa* IITR48	Oily sludge	1 h	–	Tripathi et al. [354]
Crude oil	0.5%	*P. aeruginosa* BBBE3, *P. aeruginosa* BBBJ	Soil	60 days	TPH: 95.71% (BB-BE3) and 93.28% (BBBJ)	Bhuyan and Pandey [38]
	6800 ppm/8500 ppm	*P. aeruginosa* SR17	–	3 months	TPH: 86.1% (6800 ppm) and 80.5% (8500 ppm) Floranthene/benz(b) fluorene/benz(d)anthracene: 100%Indene/chamazulene/ naphthalene/phenanthrene/ anthracene/fluorene: 60–80%	Patowary et al. [274]
	1%	*P. aeruginosa*	Soil	21 days	TPH: 91%, $C_8\sim C_{16}$: ~ 100%, $C_{17}\sim C_{44}$: ~ 60%	Rehman et al. [301]
	2%	*P. aeruginosa* PG1	Soil	35 days	TPH: 81.8%, Naphthalene: 74.67%, Fluorene: 73.37%, Phenanthrene: 70.79%, Anthracene: 68.37%, 3–3-betaMyristoylolean-12en-28-ol/1H-Indene: 100%	Patowary et al. [271]
	3%	*P. aeruginosa* SNP0614	Oil field	20 days	–	Liu et al. [208]
	10%	*P. aeruginosa* DN1	Soil	14 days	$C_8\sim C_{40}$: > 85%	Li et al. [204]
	24.98 mg/kg	P. aeruginosa	Soil	275 days	TPH: 70.54%	Khazaei et al. [182]
	2%	*P. aeruginosa* IITR48	Soil	45 days	67.1% (phenanthrene) and 61.2%(benzo(b)fluoranthene)	Kumari et al. (2018)
	–	*P. aeruginosa* P6	–	–	71.42% (pH 7.2), 27.19% (pH 5.0), 34.78% (pH 8.5)	Wang et al. [373]
Diesel oil and crude oil	2%	*P. aeruginosa* B2	Soil	35 days	TPH: 80.92%	Sharma et al. [324]
	1%	*P. aeruginosa* AKS1	Sediments	28 days	TPH: 36%	Sharma et al. [324]
	2 mL of a 3:2 mixture of diesel oil and crude oil	*P. aeruginosa* P43 and P69	Soil and groundwater	120 h	87.4% (P43) and 62.8% (P69)	Chettri et al. [69]

Table 1 (continued)

Substrate	Substrate composition	Strain	Source	Decomposition duration	Rate of removal	References
Oily sludge	2%	P. aeruginosa	Soil	56 days	$C_8 \sim C_{36}+$: 91.45%	Varjani et al. [359]
	42.25%	*P. aeruginosa NY3*	Soil	9 days	TPH: 86.2%	Luo et al. [214]
Oily waste	3%	*P. aeruginosa* NCIM 5514	–	56 days	$C_8 \sim C_{36}+$: 75.18%	Varjani and Upasani [358]
PAHs	–	*P. aeruginosa* PF2	Soil	6 h	> 99%	Pourfadakari et al. [285]
	–	*P. aeruginosa* S5	Coking wastewater	15 days	43.79%	Sun et al. [345]
	25 mg/L	*P. aeruginosa* RS1	Tank bottom sludge	21 days	≥ 95%	Ghosh and Mukherji, [124]
Naphthalene/crude oil	Naphthalene: 200 mg/L, crude oil: 1%	*P. aeruginosa* NAPH6	Seawater	7 days	Naphthalene: 99%, $C_{13} \sim C_{24}$: 96.2%	Hentati et al. [143]
Weathered hydrocarbons	TPH: 39,000–41,000 mg/kg	*P. aeruginosa* TPHK-4	Soil	90 days	TPH: 63%	Ramadass et al. [297]
Petroleum hydrocarbons	–	*P. aeruginosa* NBRC 3080	Biological Resource Center (NBRC), Japan	14 days	100% (hexacontane, hexadecane, 2,6,10,14-tetramethyl, octacosanol, octadecane and tetratriacontane)	Purnomo et al. [288]
n-C_{16}	20,000 mg/kg	*P. aeruginosa*	–	60 days	TPH: 95.7%	Hajieghrari and Hejazi, [135]
	2%	*P. aeruginosa PU1*	Soil	10 days	100%	Domdi et al. [94]
Heavy metals and petroleum hydrocarbons diesel	TPH: 3800 mg/kg	*P. aeruginosa ATCC® 9027*	–	90 days	TPH: 68%	Agnello et al. [6]

adaptability to various environmental conditions [158]. Severe environmental conditions compel bacteria to undergo a dormant phase, during which they practice adaptations such as motility improvement and genomic mutation and form endospores to paralyze growth [305]. Biosorption, bioaccumulation, and bioprecipitation are the predominant mechanisms employed by the genus *Bacillus* for the extraction of heavy metals [280]. The biosorption process is considered passive and occurs independently of biological metabolism, potentially linked to the propensity of heavy metals for adsorption onto bacterial cell wall surfaces [287]. Researchers have noted that the efficacy of biosorption is contingent upon various physicochemical parameters, including pH, starting metal level, and contact duration [91]. The processes governing biosorption differ in terms of quantity and quality depending on the species employed, the biomass source, and its treatment. The biosorption processes encompass (i) ion exchange, (ii) complexation, and (iii) physical adsorption due to intermolecular contacts [254].

Deinococcus radiodurans

Uranium, a naturally occurring radionuclide, is a common radioactive pollutant in soil, sediment, and groundwater, with three main isotopes: 238U, 235U, and 234U [185]. Uranium, a long-lived metal, poses significant environmental and health risks when ingested through food and water, potentially damaging the liver, kidneys, nervous system, and immune system and leading to cancer [56]. Chemical properties and radioactivity determine uranium toxicity, with tetravalent and hexavalent forms found in radioactive wastewater. Sedimentary U(IV) compounds have low toxicity, while soluble U(VI) has high toxicity [336]. The primary objective of bioremediation for radioactive uranium wastewater is the removal of hexavalent uranium from the effluent [395].

Deinococcus radiodurans is a microorganism that can handle radiation the best. It is a Gram-positive bacterium that is not harmful to humans and is very resistant to ionizing radiation, UV light, drying out, mitomycin C, and oxidative stress (Sadraeain & Molaee 2009). These bacteria act as sponges, absorbing radionuclides from the environment with minimal harm to themselves due to complex biological systems that effectively safeguard their proteins from radiation damage [193]. *Deinococcus radiodurans* reduces the bioavailability and associated risks of hazardous radioactive substances by absorbing them, thereby cleansing the environment [198, 201, 202]. *Deinococcus radiodurans* actively contribute to the remediation of radioactive-contaminated sites, facilitating the restoration of impacted areas and protecting both human health and the environment from the adverse effects of radioactive pollution [199, 203, 205].

***Sphingomonas* spp.**

Polycyclic aromatic hydrocarbons (PAHs) are deemed harmful owing to their toxic, genotoxic, mutagenic, and/or carcinogenic characteristics [267, 269]. Microbial removal is a critical mechanism for the effective removal and eradication of PAHs from polluted soils [350]. Native or indigenous microbiota play a crucial role in the bioremediation of petroleum-contaminated sites, often outperforming commercial inocula due to their potential to outcompete them [236]. Over the past 2 decades, scientists have been studying microorganisms' potential environmental benefits, including bioremediation, crop plant growth promotion, and secondary metabolite production [311]. Habitat-specific symbiosis and microbe genetic makeup contribute to beneficial properties, with rhizobacteria being particularly notable for their role in plant growth promotion, stress tolerance, and crop production [191]. The bacterium genus *Sphingomonas* is known for its proactive role in environmental remediation [180]. *Sphingomonas* are Gram-negative bacteria that come from more than 103 species. They look like yellow rods that do not move and have a single polar flagellum [125]. They play a crucial role in plant abiotic stress tolerance [212], bioremediation, and environmental contaminants removal [410]. *Sphingomonas* species exhibit unique aromatic hydrocarbon removal pathways, promoting environmental bioremediation of heavy metals and PAHs with distinct gene organization compared to other *Pseudomonas*, *Ralstonia*, and *Burkholderia genera* [20] Table 2.

Zhou et al. [410] proposed the genus *Sphingomonas*, which often isolates PAH-contaminated soils and likely plays a crucial role in natural PAH-degrading microbial consortiums. Several investigations indicated that *Sphingomonas* constituted one of the principal groups within the microorganisms of PAH-contaminated water and soil specimens [223]. In addition, *Sphingomonas paucimobilis* may serve as a secure biological remediation technique for soil contaminated with hydrocarbons at crude oil extraction sites [164].

***Rhodococcus* spp.**

Toxic, mutagenic, teratogenic, and carcinogenic contaminants are harmful to plants, animals, and people because they affect cell growth and the immune, respiratory, reproductive, endocrine, and neurological systems [266, 298]. Researchers have refined treatment methods to effectively eliminate or remove pollutants, thereby reducing associated dangers [248]. Researchers have advanced treatment methods that effectively eliminate or remove pollutants, thereby reducing the related dangers [248].

Actinomycetes are Gram-positive bacteria with significant potential for use in bioremediation procedures

Table 2 Bacterial bioremediation mechanisms

Mechanism	Description	Example	References
Adsorption	Bacteria bind heavy metals on their cell surfaces using functional groups such as carboxyl, hydroxyl, and phosphate	*Bacillus cereus* adsorbs Pb(II), Cr(VI), and As(V) using exopolysaccharides	[400, 404]
Bioleaching	Bacteria produce organic acids or enzymes to solubilize metal ions, making them more bioavailable for removal	*Cupriavidus metallidurans* bioleaches cadmium and mercury while degrading benzene	[18]
Biomineralization	Bacteria convert heavy metals into stable mineral forms to reduce toxicity and mobility	*Bacillus* species precipitate cadmium and lead as stable complexes in contaminated soils	[400, 404]
Biotransformation	Bacteria alter metal oxidation states, reducing toxicity and enhancing removal	*Halomonas* sp. oxidizes arsenite (As^{3+}) to the less toxic arsenate (As^{5+})	[400, 404]
Cellular deposition	Bacteria accumulate heavy metals inside their cells to prevent toxicity	*Vibrio harveyi* stores cadmium ions up to 23.3 mg Cd^{2+}/g dry biomass	[1]
Enzymatic oxidation	Bacteria use enzymes to oxidize pollutants, breaking down toxic compounds	*Pseudomonas aeruginosa* utilizes laccase to degrade polycyclic aromatic hydrocarbons (PAHs)	[117]
Enzymatic reduction	Bacterial reductases convert harmful metals to less toxic, insoluble forms	*Bacillus cereus* uses chrome reductase to reduce Cr(VI) to Cr(III)	[317]
Efflux pump mechanism	Bacteria expel heavy metals using membrane transport systems to survive in contaminated environments	*Delftia lacustris* resists Cr(VI), Hg(II), Cd(II), and Pb(II) by using efflux pumps	[381]
Exopolysaccharide (EPS) production	Bacteria secrete EPS to bind and immobilize heavy metals, reducing bioavailability	*Bacillus zhejiangensis* removes 90% of Cd^{2+} and 91% of Pb^{2+} through EPS-mediated biosorption	[144]
Consortium-based bioremediation	Mixed bacterial communities work together to degrade pollutants efficiently	A bacterial consortium effectively removes mercury ions from contaminated environments	[292]
Hydrocarbon degradation	Bacteria break down petroleum hydrocarbons using metabolic pathways	*Pseudomonas aeruginosa san ai* degrades n-alkanes and PAHs with up to 98% efficiency	Medić et al. [233]
Bioaugmentation	Introducing specific bacteria to contaminated sites to enhance bioremediation	*Pseudoxanthomonas mexicana* transforms and removes mercury and arsenic in wastewater	[3]
Biostimulation	Optimizing environmental conditions (e.g., pH, oxygen, and nutrients) to enhance bacterial activity	*Enterobacter kobei* increases lead removal efficiency under optimized conditions	[105]

[299]. *Rhodococcus*, a genus of actinomycetes, has significant potential for treating various organic and inorganic compounds, including PAHs, phenolic compounds, fungicides, pharmaceuticals, phthalates, steroids, dyes, and heavy metals [32, 163, 299]. Moreover, several materials have immobilized *Rhodococcus* to aid in the biosorption of pollutants from liquid and gaseous phases [250]. *Rhodococcus* strains' possibility of bioremediation is attributed to their unique enzyme systems and high resistance against ecotoxicants, making them suitable for various applications [42].

Cupriavidus metallidurans

Bioremediation is a cost-effective and long-term remediation strategy using biological systems to reduce heavy metal concentrations through sorption, biofilm accumulation, and uptake [75]. A multitude of microorganisms pose challenges for isolation and ex-situ maintenance, necessitating the development of protocols for their growth and maintenance [171]. An example is *Cupriavidus necator*, which exhibits significant resistance to heavy metals and demonstrates enhanced growth in the presence of Cu [382]. The genus *Cupriavidus* is in the class β-Proteobacteria and the family *Burkholderiaceae*. It is composed of Gram-negative, aerobic rods that use petrichorous flagella to propel themselves and develop smooth colonies of 1–2 mm in diameter within 48 h at 30 °C [78]. Cupriavidus species flourish in metal-rich habitats, particularly in Cu-laden soils characterized by a combination of toxic transition metal cations, resulting in elevated cellular Cu levels. We still do not fully understand how *Cupriavidus* spp. affects copper partitioning in the environment. The commencement of growth in *Cupriavidus* spp. has been shown to be enhanced by Cu [382]. Their capacity to assimilate Cu and consume other microbes confers a competitive advantage in the ecosystem [340]. *C. necator* UFLA 01–659 has been demonstrated to absorb all tested heavy metals on the cell wall and membrane, while complexation was exclusively detected intracellularly for Cu and Zn [363]. In addition, *Cupriavidus metallidurans* has demonstrated the ability to repair mercury in contaminated agricultural soil and different metal ions, including lead and cadmium [45].

Mycobacterium sp.

Polycyclic aromatic hydrocarbons have emerged as a considerable environmental issue owing to their potential toxicity, mutagenicity, and carcinogenicity [267, 269]. PAHs are broken down by dioxygenase enzymes adding two oxygen atoms to PAH molecules [385]. *Mycobacterium species* are the only bacteria that can do this. *M. vanbaalenii* PYR-1 was initially demonstrated to break down low-molecular-weight compounds such as 4-ring pyrene [276]. The distinctive cell wall structure of mycobacteria, characterized by mycolic acids, is crucial for the mycobacterial breakdown of hydrophobic polycyclic aromatic hydrocarbons [365]. Pyrene-degrading mycobacteria may metabolize several PAHs, such as "naphthalene, phenanthrene, and fluoranthene, as well as five-ring molecules like benzo[α]pyrene, through co-metabolic removal", utilizing these compounds as carbon sources [16].

Zeng et al. [397] demonstrated that the PdoAB dioxygenase from *Mycobacterium* sp. NJS-P performs diverse activities in the oxidation of several PAHs, including molecules with up to five rings, such as benzo[α]pyrene. In addition, pdoAB can be activated during bacterial proliferation on pyrene and phenanthrene, indicating its role in PAH breakdown. *Mycobacterium* species in rhizosphere soils have been demonstrated to expedite the breakdown of organic pollutants such as PAHs and improve plant resilience to soil contamination [82]. In addition, the pyrene-degrading bacterium *Mycobacterium* sp. Pyr9 displayed several plant growth-promoting traits and significant resilience to extreme conditions, indicating substantial promise for bioremediation applications [388]. Consequently, pyrene-degradative Mycobacterium strains may improve the security of agricultural products and human wellness in regions affected by PAHs [90].

Alcaligenes eutrophus

Global soil environment contamination by petroleum hydrocarbons is increasing due to heavy petroleum dependence, rapid industrialization, population growth, and disregard for environmental health [355]. Hydrocarbons, whether accidentally released or induced by humans, disrupt natural ecosystems, disrupt species equilibrium, and pose significant threats to living species as carcinogens and neurotoxic organic pollutants [200]. Aromatic hydrocarbons are prevalent environmental contaminants possessing toxic, genotoxic, mutagenic, and carcinogenic characteristics [364]. Pipeline breaks, tank leaks, and storage accidents in the petroleum industry often let hydrocarbons like PAHs into the soil. BTEX compounds are known to cause cancer and damage neurons [239].

There is a big role for bacteria in breaking down hydrocarbons. Low-molecular-weight alkanes break down the fastest, but mixed cultures break down more petroleum than pure cultures [26]. Removal using microorganisms is preferred for PAHs removal due to its cost-effectiveness and comprehensive cleanup advantages [244]. Reports have indicated the potential applications of *Alcaligenes faecalis*, *Alcaligenes eutrophus* A5, and *Alcaligenes aquatilis* LMG 22996T in agriculture, pharmaceutical industries, and pesticide removal [209]. *Alcaligenes aquatilis* QD168 breaks down marine hydrocarbons, and *A. aquatilis* F8 strain makes cationic biosurfactants. These microbes may be able to adapt to environmental stressors such as toxic compounds and high salinity [98]. These attributes indicate the strain's potential for bioremediation of oil-contaminated environments [221].

Shewanella oneidensis

Oil spills primarily cause hydrocarbon contamination in marine environments. The procedure of degassing and marine extraction of crude oil mostly results from partial combustion of organic components such as timber and petroleum [341]. Bioaugmentation, which involves adding microorganisms that can break down things, has been suggested as an alternative to chemical and physical therapies because it is cheaper and has less of an impact on the environment [393]. Researchers have tested bacterial strains from contaminated soils for mineralizing low-molecular-weight hydrocarbons and biodegrading PAHs, but few studies have focused on marine sediment bioremediation [39]. Furthermore, Cai et al. [49] demonstrate that RNAP mutations effectively screen bioremediation candidates, clarify gene co-expression networks, and expand the application of *Shewanella* spp. in azo dye removal.

Shewanella spp. can donate electrons in several different ways, which makes them useful for getting rid of contaminants such as nitrate, TMAO, azo dyes, and oxidized metals. They thrive in high salinity and heavy metal environments [282]. It has been seen that *Shewanella* spp. can break down solid oxidized metals in used LIBs, such as Ni (IV), Co (IV), and Mn (IV). Cathodic substances may liberate these metals into the growing medium [49]. Another study used unbiased RNA polymerase mutation screening to find Shewanella strains that are very good at bioremediation, changing a lot of different phenotypes and worldwide transcriptional patterns [50].

Synergistic bacterial consortia for enhanced bioremediation

Microbes found in various environments play a crucial role in global biogeochemical cycles, impacting human survival, health, and development [400, 404]. Some microbes isolated in laboratories could remove pollutants and transform heavy metals [264]. Nonetheless, the pollutant transformation efficiency of a single species consistently diminishes when utilized in complex, in-situ contaminated environments [307]. Complex pollutants cause stress and hinder metabolism in single species, while microbial consortiums exhibit adaptation and versatility are crucial as many species effectively exploit substrates [380].

Engineering collaborations between microbes may effectively optimize interactions among microorganisms and their surroundings to guarantee enduring stability. Communities of bacteria exhibit competition for scarce nutrients and utilize metabolic byproducts excreted by other species to enhance their fitness [293]. Metabolic interactions significantly influence the application of microbial consortia, with substances secreted by specific species affecting growth, interactions, and community function,synthetic microbial consortia are artificially created [206]. In contrast to organic microbial collaborations, synthetic microbial consortia can have their metabolic pathways reengineered and social relationships orchestrated to achieve the required functionality [97].

Unlike monocultures, consortia can metabolize a broader range of contaminants through cooperative metabolic pathways, increasing efficiency and adaptability to environmental stressors. One notable example is the use of bacterial consortia supplemented with rhamnolipids for petroleum hydrocarbon degradation. This approach significantly enhanced total petroleum hydrocarbon biodegradation, achieving an 81% removal rate within 35 days. The synergistic interaction between *Pseudomonas*, *Pseudoxanthomonas*, *Cavicella*, *Mycobacterium*, *Rhizobium*, and *Acinetobacter* altered bacterial community composition and increased the abundance of hydrocarbon-degrading genes, demonstrating a promising strategy for bioremediation of oil-contaminated soils [384]. Similarly, consortia of *Mycobacterium*, *Rhodococcus*, and other PAH-degrading bacteria have shown high degradation efficiencies of up to 60% for high-molecular-weight PAHs when combined with strategies such as nutrient supplementation and microbial immobilization on biochar. This combination significantly improved the degradation of aged PAHs in contaminated soils, making it a viable approach for long-term bioremediation projects [408, 409].

Moreover, bacterial–fungal consortia have proven effective in the simultaneous removal of heavy metals and organic pollutants. A specific microbial consortium derived from activated sludge and copper mine sludge achieved 95.6% degradation of phenanthrene within 7 days, while demonstrating a high tolerance to cadmium (Cd^{2+}) contamination. When immobilized on biochar, the consortium improved bioavailability and enzymatic activity, leading to near-complete removal of phenanthrene and cadmium [199, 203, 205]. In heavy metal bioremediation, indigenous bacterial consortia composed of *Aeromonas caviae*, *Aeromonas hydrophila*, and *Shewanella putrefaciens* demonstrated superior resistance and removal efficiencies for Cu (47.02%), Ni (61.49%), and Zn (61.93%) compared to monocultures. The synergistic interactions between these bacteria enabled multiple metal removal mechanisms, including bioaccumulation, biosorption, and biotransformation, making them ideal candidates for heavy metal remediation [183]. Another effective approach involves immobilized bacterial consortia for remediating intertidal zones contaminated with heavy oil. In a 100-day experiment, immobilized consortia achieved a 52.99% degradation efficiency, significantly outperforming natural attenuation and biostimulation. Gas chromatography–mass spectrometry (GC–MS) analysis revealed that *Acinetobacter* and *Bacillus* species played a dominant role in the degradation of C15–C35 n-alkanes and PAHs, demonstrating their potential for large-scale coastal bioremediation applications [81].

In addition, engineered microbial consortia have been developed to tackle complex hydrocarbon pollution. A novel approach using a five-strain bacterial consortium achieved a 31.54% higher degradation rate than single-strain cultures, emphasizing the stability and efficiency of multi-strain bioremediation strategies [66]. A consortium composed of *Acinetobacter* and *Rhodococcus* in combination with nitrogen-fixing *Azotobacter vinelandii* removed 83.1% of diesel contamination without requiring external nitrogen supplementation. This approach reduces dependency on chemical fertilizers, making it a sustainable and cost-effective bioremediation strategy [60]. In addition, microbial consortia have been optimized for treating lubricant-contaminated sites containing both hydrocarbons and heavy metals. A consortium of *Lysinbacillus fusiformis* and *Bacillus tropicus* successfully removed 90.3% of lubricating oil and 84.2% of Cr(VI) within 6 days. The bacterial response to chromium exposure induced the secretion of extracellular protein-like substances, aiding in metal detoxification and organic pollutant degradation [151, 153].

The design of microbial partnerships is a difficult endeavor that entails the assembling of several genera and species of microorganisms. Two primary techniques and numerous key concepts govern the engineering of microbial consortia for bioremediation [198, 201, 202].

Top-down engineering

Top-down engineering employs a natural microbial population to execute certain activities by altering ecological factors, including temperature, pH, average incubation durations, salinity, redox conditions, and carbon sources [196]. It offered an efficient method for bioremediation. A research investigation demonstrated that the inclusion of electron donors facilitated the proliferation of microbial consortia and enhanced the breakdown of harmful chlorinated pollutants [150]. For example, reactor engineering or biostimulation in wastewater treatment plants can help key species grow. Still, top-down engineering focuses on self-assembling microbial consortia without taking species specificity or metabolic pathways into account [293]. Top-down engineering requires manipulating microbial consortiums, and modeling helps predict the desired function. Physicochemical conditions and abiotic and biotic processes influence biological processes. Capturing stoichiometric and kinetic parameters improves efficiency and control of bioremediation costs [216]. Top-down engineering is an effective method for ecological modification and has limitations due to its inability to accurately control microbial consortia and their complex metabolism accurately, hindering system optimization [170].

Numerous contaminated locations are simultaneously affected by organic and metallic contaminants, resulting in a cumulative stress effect on microorganisms [127]. By affecting the enzymes used in biological remediation or metabolism, metal toxicity can affect the removal of organic substances [46]. Furthermore, metals such as Cr (VI) can inflict harm on cytoplasmic membranes or DNA [290]. Likewise, the existence of organic contaminants exerted stress on microorganisms. PAHs such as phenanthrene (PHE) and pyrene (PYE) exhibit significant toxicity to cell membranes, leading to membrane malfunction or instability [121]. Furthermore, PAHs may modify metal buildup and increase metal-derived reactive oxygen species (ROS) [89]. Therefore, mitigating stress caused by intricate pollution is crucial for the advancement of top-down engineering.

Bottom-up engineering

Multi-omics and automation methods have revolutionized bottom-up engineering by enabling researchers to control metabolic pathways and interactions between bacteria, overcoming challenges in predicting and managing microbial consortiums [216]. Metabolism determines a species' nutrient consumption and excretion. Pathway models such as glycolysis, Kreb's cycle, and glutaminolysis help predict output. Genome sequences and quantitative models are used to reconstruct metabolic pathways and investigate microbial consortia dynamics [109]. The formation of organisms with defined social relations is an essential component of bottom-up engineering. Microbial relationships, encompassing rivalry as well as cooperation, are ubiquitous in communities of microbes and are essential in shaping the dynamics of ecosystems [307]. A model utilizing social-interaction programming may effectively forecast the actions and interactions within an ecosystem of up to four types [186]. Bottom-up engineering allows for the creation and optimization of microbial consortiums for bioremediation of complex pollutants, but challenges arise due to inaccurate or incomplete data [198, 201, 202].

Pollutant removal pathways are complex, with microbes having varied metabolic mechanisms and resource preferences, making it energy-efficient for a particular kind of execution of the process independently. Toxic intermediates inhibit removal [54]. It is essential to establish a specialization of tasks within metabolic pathways for the removal of certain contaminants. The metabolic division of labor makes bioremediation easier because one organism may be better at breaking down pollution at a certain stage. Synthetic pyrene-degrading consortia are an example of this [367]. Mycobacterium strains possess an extensive gene repertoire for pyrene breakdown, yet they cannot utilize it. *Novosphingobium* and *Ochrobactrum* effectively break phthalate or protocatechuate, contributing to pyrene intermediate removal [194]. The species that are best at dividing their metabolisms help microbial groups break down pollutants quickly and cheaply, but only one species breaks down atrazine directly (Fig. 1).

Microbes form multifunctional communities in the natural environment, with diversity contributing to their stability (Hampson et al. 2019). These communities have enhanced access to nutrients and support, allowing them to withstand environmental disturbances and improve their functions. For instance, *Salmonella* cannot directly utilize lactose, but *Escherichia coli* can metabolize it [137]. The surfactant-making function of microbial consortiums is important for low-water-soluble organic pollutants because it provides nitrogen fixers and carbon producers for bioremediation. *Bacillus* strains can enhance pyrene bioavailability by producing biosurfactants [367]. A group of microbes called *Pseudomonas* and *Actinobacteria* strains break down PAHs. These microbes have emulsifying properties that help break down PAHs [159]. Synthetic microbial consortia, primarily responsible for degrading hazardous substances, also include other partner species with diverse functions, which can enhance bioremediation in the process (Fig. 2).

Applications in different environmental settings

Marine oil spills

The seas, oceans, and coastal regions are under considerable strain, with pollution, particularly from crude oil, presenting a serious risk to the planet's sustainability [322]. Annually, around 1.3 million tons of oil permeate the marine ecosystem [65]. Acute pollution events elicit significant public anxiety, particularly the approximately "600,000 tons of crude oil discharged during the Deepwater Horizon explosion in the Gulf of Mexico" [77] and around "63,000 tons from the Prestige oil tanker off the northwest coast of Spain" [175]. The outcome of crude oil discharged into the ocean is influenced by the existing weather conditions and the oil's composition,nonetheless, its ecological consequences intensify upon contact with the shoreline, particularly in low-energy environments such as lagoons and salt marshes (Fig. 3). The Prestige spill resulted in the loss of over 66% of total species richness in the worst affected beaches, highlighting the devastating impact of acute pollution events [21]. Hydrocarbons pollute the plumage and pelage of marine avifauna and mammals, resulting in a deterioration of hydrophobic characteristics, hypothermia, or lethal doses after oil consumption during preening [353].

Hydrocarbons, particularly PAHs, can have long-lasting impacts on wildlife and fisheries, causing physiological and behavioral damage, genetic damage, and decreased growth and fecundity in fish [146]. Drilling chronically impacts deep-sea sediments and biota, resulting in oil-contaminated drilling cuttings. Toxic components, like high-molecular-weight PAHs, can remain buried and released into the environment [147]. Hydrocarbons, stable molecules with a long-standing presence in the environment, have evolved into microbes that can utilize them as a principal supply of both energy and carbon [72]. Haloarchaeal genera and Eukarya can change hydrocarbons. One important way to clean up environments polluted with hydrocarbons is for crude oil to break down into carbon dioxide and water through removal [232].

Factors such as oil composition, weathering, temperature, and nutrient concentrations influence the microbiological response to a marine oil spill. Nonetheless, certain usual patterns exist; most prominently, there exists a substantial rise in the prevalence of *Alcanivorax* spp., which metabolizes straight-chain and branched alkanes [188], succeeded by *Cycloclasticus* spp., which breakdown PAHs [251]. *Alcanivorax borkumensis* has a limited range of catabolic pathways, but it does have several alkane-catabolism pathways involving essential enzymes such as alkane hydroxylases and cytochrome P450-dependent alkane monooxygenases [128]. The type of alkane and growth phase influences the expression of *Alcanivorax borkumensis* and *Acinetobacter* spp. They adapt to access oil, survive in open marine environments, and have diverse alkane hydroxylase systems [176]. Oil spills in cold marine environments often involve *Oleispira*, an obligate alkane-degrading psychrophile that outperforms *Alcanivorax* spp. in temperate conditions and makes up 90% of the global microbial community [28]. Generalists,

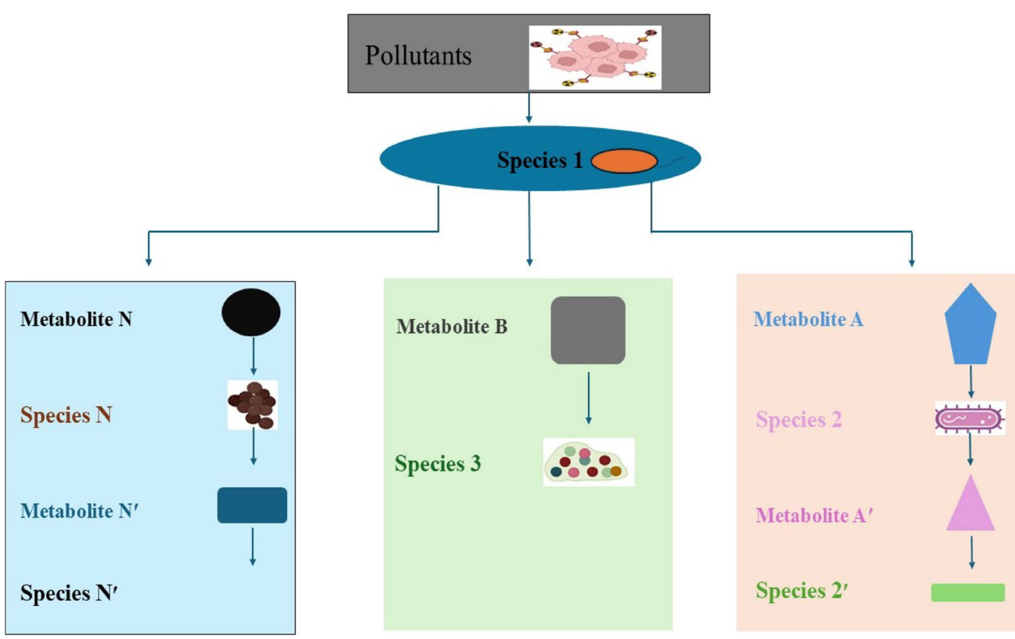

Fig. 1 The allocation of tasks within a metabolic circuit

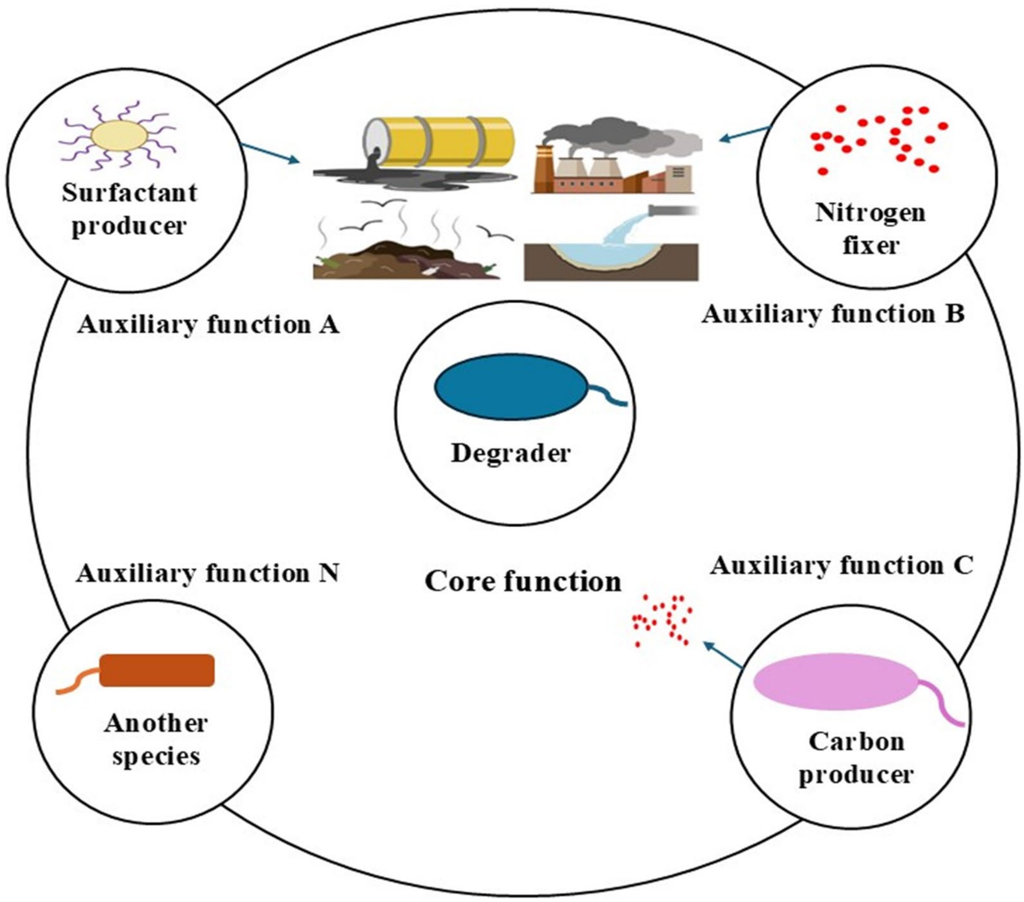

Fig. 2 A microbial community exhibits multifunctionality. The removal of contaminants is a fundamental function of a synthetic microbial community. Auxiliary tasks, such as the production of surfactants, nitrogen, or carbon resources, assist the removers in fulfilling their primary role

including "*Roseobacter*, *Acinetobacter*, *Marinobacter*, *Pseudomonas*, and *Rhodococcus* spp.," contribute significantly to hydrocarbon removal in marine waters. These microbes, often found in the aerobic zone of marine sediments, play a crucial role in the hydrocarbon-degrading community [12].

Heavy metal contamination

Heavy metals pollution in agriculture poses a significant obstacle to developing and poor nations. Remediation using microorganisms or biocatalysts involves integrating genomics, understanding transcriptomics, proteomics, communication structures, and artificial biology is essential [347]. HMs impact soil microbiology, modifying plant–microorganism connections and influencing soil properties, growth, flora, and agricultural yield. Distinct microbial populations possess unique metabolic capabilities, forming organic substances or reducing HMs [398]. Industries release toxic HMs, impacting soil fertility, microbial functions, and agricultural yield. Polluted foods present significant health hazards to humans [253]. Global agricultural communities are concerned about soil contamination with toxic HMs, including cadmium, lead, zinc, copper, arsenic, and mercury, which can persist over time through organic and inorganic fertilizers [131].

Microbes interact with HMs through various mechanisms, influenced by factors such as climate, pH, nutrition origin, and ions of metals. The diminutive size, swift proliferation, and straightforward cultivation of bacteria facilitate their success in diverse situations [17]. Different bacteria can take in different amounts of HMs. For example, the Hg-resistant *Pseudomonas aeruginosa* strain can take in 180 mg/g. *Bacillus* sp. PZ-1 and *Pseudomonas* also absorb Pb from wastewater, while *Arthrobacter viscosus* can absorb and transfer Cr (VI) to Cr (III) [347]. Bacillus cereus and *Rhodobacter capsulatus* can take in different amounts of Zn (II) and Cd (II) in cells [219]. Metal ions such as Cd, Hg, Cu, and cobalt cannot get into microorganisms because of extracellular polymeric substances (EPS) that block their entry and store them (Fang et al. 2017). *Pseudomonas putida* is a crucial microbe

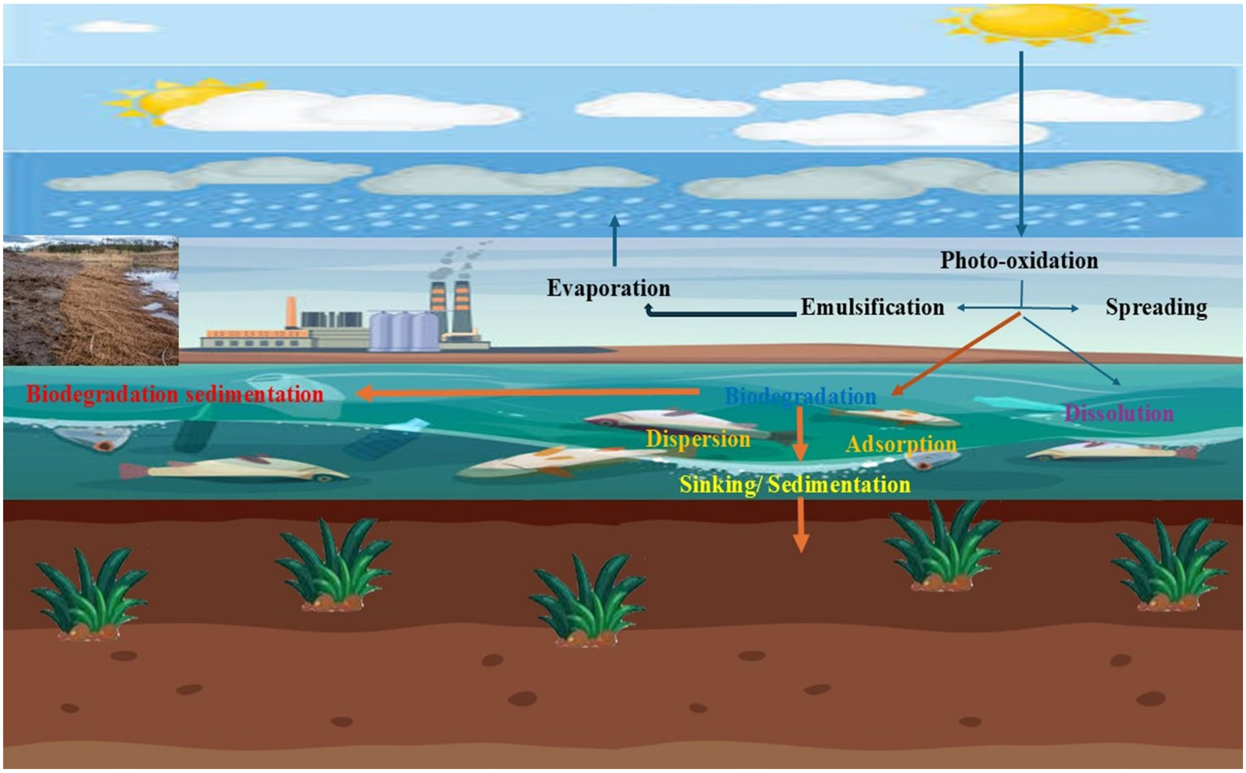

Fig. 3 Outcome of a marine oil spill. Factors such as winds, waves, water currents, oils type, and temperature influence the spread of oil and can increase the evaporation of volatile fractions. Turbulent seas can cause water-in-oil emulsions, which are difficult to remove due to their high viscosity. Photo-oxidation can also increase water solubility or resistance to removal. Sedimentation occurs when oil adsorbs to particles

capable of absorbing 100% Hg from the marine ecosystem and reducing Hg (II) into Hg (0) [403].

The amount of mercury in polluted soils dropped by a lot when the pTP6 plasmid was added to genetically modified *Cupriavidus metallidurans* [93]. The addition of new genes to *Pseudomonas* using the pMR68 plasmid led to the development of mercury resistance [335]. Changing the genes of microbial membrane transporters can help remove heavy metals from soils more effectively since transporters and binding processes are necessary for this to happen [227]. Several phytochelatins, metallothioneins, and polyphosphates collaborate within cells to absorb heavy metals and alter their primary storage mechanism. This makes it easier for cells to take heavy metals from soil and water [92].

In addition to conventional bioremediation strategies, recent advances have demonstrated the effectiveness of genetically engineered bacteria in heavy metal remediation. For instance, a study employing *Bacillus subtilis* as a chassis organism successfully engineered the bacterium to sense and remove Pb^{2+}, Cu^{2+}, and Hg^{2+}, achieving metal removal efficiencies above 99%. The engineered *Bacillus subtilis* biofilm also enhanced plant growth when combined with biochar, demonstrating its potential for soil and water remediation [412]. Similarly, *Bacillus cereus* and *Enterobacter cloacae* strains isolated from industrial wastewater exhibited strong tolerance to Cr, Pb, and Ni, with *Bacillus cereus* showing over 60% Pb removal efficiency under optimized conditions [178]. In addition, studies have reported that *Burkholderia cepacia*, a heavy-metal-tolerant soil bacterium, enhances cadmium (Cd) and lead (Pb) removal while also acting as a plant growth promoter by solubilizing phosphate and producing siderophores [167]. A study of metal-resistant strains isolated from contaminated environments has shown great promise for HM remediation. *Stenotrophomonas rhizophila*, a bacterium isolated from tannery effluent, demonstrated high resistance and degradation efficiency for arsenic (91.6%), mercury (91%), and chromium (78.66%). This bacterium also exhibited plant growth-promoting properties, making it a valuable candidate for remediating heavy-metal-polluted soils in agricultural settings [260].

Furthermore, biosurfactant-producing bacteria such as *Bacillus subtilis* and *Paenibacillus dendritiformis* have been utilized for co-contaminated site remediation, targeting both hydrocarbons and heavy metals [107]. A study of thermophilic bacteria was conducted in the Western Garhwal Himalayas isolated *Pseudomonas alloputida* and *Paenibacillus thiaminolyticus*, which achieved over 50% Cu^{2+} removal efficiency. Interestingly, dead-cell biomass of these bacteria was found to be even more effective than live cells, suggesting its potential as a passive bioremediation approach for water treatment [366]. Moreover, another study investigating *Serratia quinivorans* and *Pseudomonas hibiscicola* found that these bacteria could efficiently remediate mercury-contaminated wastewater, reducing its toxicity by more than fivefold within 72 h. The application of these strains in phytoremediation systems further improved their mercury removal efficiency, demonstrating their potential for industrial wastewater treatment [386].

Pesticide contamination

The genus *Rhodococcus* was quickly recognized as one of the most attractive groups of organisms for the removal of chemicals that are not readily converted by other species [161]. *Rhodococci*, a genus of bacteria, has been found to possess comparable removal abilities to *pseudomonads* and related bacteria [55]. They metabolize various xenobiotic substances, "encompassing aliphatic and aromatic hydrocarbons, oxygenated chemicals, halogenated substances, and herbicides" [162]. *Rhodococci* had the greatest interest in the removal of hydrocarbons, then nitriles and phenolic substances [249].

Rhodococcus opacus is a Gram-positive, *chemoorganotrophic Actinomycetales* organism that does not cause disease. Its cell wall is made up of polysaccharides, carboxylic acids, lipids, and mycolic acids [165]. Actinomycetes from the genus *Rhodococcus* can break down a wide range of organic and inorganic substances, such as heavy metals, phthalates, phenolic compounds, fungicides, pharmaceuticals, phthalates, and dyes [226, 299, 405]. Furthermore, *Rhodococcus* strains' bioremediation potential is attributed to their unique enzyme systems and high resistance against ecotoxicants, making them suitable for various applications [42]. *Rhodococcus* strains exhibit superior efficacy in bioremediation and bioconversion of refractory contaminants attributable to their adaptability and tolerance to toxic substances compared to other microorganisms [406].

Organophosphate pesticides persist in the environment, posing severe health and ecological risks. However, bacteria such as *Bacillus*, *Pseudomonas*, *Acinetobacter*, *Agrobacterium*, and *Rhizobium* can effectively degrade pesticides such as profenofos, quinalphos, malathion, methyl-parathion, and chlorpyrifos, mitigating their harmful impact on ecosystems [349]. Recent studies highlight the potential of plant-growth-promoting rhizobacteria (PGPR) in remediating pesticide-heavy metal co-contaminated soils. PGPR strains can improve soil fertility, degrade pesticides, and reduce heavy metal toxicity by altering metal pathways and producing bioactive compounds such as siderophores and chelating agents [326]. Furthermore, the co-application of PGPR with biochar, organic acids, and plant growth regulators has been shown to enhance pesticide degradation while mitigating metal toxicity in agricultural soils [326]. Advancements in microbial biotechnology, including gene editing tools like CRISPR/Cas9, have enhanced bacterial degradation of organophosphate, organochlorine, and pyrethroid pesticides. Engineered strains show improved enzymatic pathways, leading to more efficient pesticide removal and reduced soil and water contamination [300].

Microbial consortia have proven highly effective in pesticide bioremediation. A six-bacteria mix, including *Bacillus subtilis*, *Bacillus pumilus*, and *Pseudomonas aeruginosa*, achieved 99.33% degradation of chlorantraniliprole in 20 days, demonstrating the potential of multi-strain approaches for rapid pesticide removal and complete mineralization of the pesticide [112]. Pesticide storage sites present serious environmental hazards, but in-situ bioremediation using *Stenotrophomonas* sp., *Lysinibacillus fusiformis*, and *Enterobacter cloacae* successfully degraded 85.6 to 99% of pesticides in Kyrgyzstan within 6 months [95]. Pyrethroid pesticides are particularly persistent, but *Rhodococcus ruber Y14* has been identified as an efficient degrader. This strain removed over 80% of D-cyphenothrin from soil within 25 days through carboxyl bond hydrolysis. This strain was able to completely degrade multiple pyrethroid insecticides, including D-cyphenothrin, allethrin, fenvalerate, tetramethrin, and prallethrin, highlighting its potential for large-scale bioremediation applications [151, 153].

Plastic removal

Plastic waste in landfills leads to pollution of the environment because of the inclusion of toxic and dangerous materials in the refuse [270]. Plastic garbage, which generates particle matter and removes under UV radiation, can readily infiltrate the food chain, possibly resulting in the mortality of living animals [62]. This results in microplastic waste, which diminishes light transmission, photosynthetic efficacy, marine creature production, and sediment deposition [156]. Bioremediation technology can treat microplastics in aquatic environments using biofilms, with seagrasses and macrophytes being suitable candidates. Primary and secondary methods produce microplastics: cosmetics, industrial cleaning, synthetic textiles, and wastewater produce primary microplastics, while natural weathering produces secondary plastics [346].

Plastic waste in aquatic environments, originating from industrial waste, marine fisheries, and atmospheric release, increases surface area and adsorption capacity, interacting with heavy metals and pathogens [338]. Marine animals can ingest or entangle plastics, which break down into microplastics due to UV radiation and microbial activities, causing fragmentation, removal, and weathering [83]. Microplastic particles accumulate on the seabed, facilitating microbial proliferation and biofilm development, which enhances the breakdown of microplastics by marine microorganisms [371].

Lipases, cutinases, and esterases are some of the enzymes that can break down PET. They can do this at high temperatures (50–70 °C), but not very well at room temperature [289]. In addition, Yoshida et al. [394] isolated the bacteria *Ideonella sakaiensis* 201-F6 from a waste recycling facility in 2016. The bacteria may secrete PET hydrolase (PETase) and MHETase to decompose PET into intermediate products at 30 °C, thereby laying the groundwork for removal at ambient temperatures. After that, scientists looked at the structures of PETase and MHETase and did several studies on how to change enzymes in a way that worked well, which led to important findings [375].

Removal of antibiotics and pharmaceutical pollutants

The increasing contamination of water bodies with antibiotics and pharmaceutical residues due to pharmaceutical discharge and agricultural runoff poses a serious environmental threat. Conventional wastewater treatment processes are often insufficient in removing these contaminants, necessitating alternative remediation strategies. Microbial degradation has shown potential, with bacterial consortia capable of metabolizing pharmaceutical pollutants under metabolic and co-metabolic conditions [7, 11]. Bacterial species such as *Pseudomonas*, *Bacillus*, and *Acinetobacter* have demonstrated the ability to degrade antibiotics through enzymatic pathways, mitigating their ecological impact [14, 179]. *Bacillus subtilis* strains, for example, produce β-lactamases that can degrade β-lactam antibiotics such as amoxicillin and cephalexin, achieving degradation rates of up to 25% under optimal conditions [14, 15].

In addition, lipopeptides produced by *Pseudomonas* and *Bacillus* play a crucial role in biofilm formation and microbial competition, which enhances bioremediation efficiency [184, 291]. The presence of antimicrobial resistance genes (ARGs) in wastewater treatment plants is a growing concern, as ARGs can be horizontally transferred among microbial communities, further exacerbating antibiotic resistance [179, 379]. The use of bacterial consortia in wastewater treatment plants presents a promising approach to mitigating ARG dissemination, particularly when combined with advanced bioremediation technologies such as bioaugmentation and nanotechnology [215, 321]. Recent studies have demonstrated that antioxidant-producing microorganisms can mitigate ARG enrichment and dissemination by reducing ROS levels, thereby limiting genetic mutations and plasmid-mediated ARG transfer in microbial communities [87, 302]. In parallel, bioremediation using extremophiles—microorganisms adapted to extreme environmental conditions—has emerged as a promising approach for pollutant degradation [113, 318]. Extremophilic bacteria and their specialized enzymes exhibit high metabolic versatility, enabling them to degrade pharmaceutical contaminants, including antibiotics, in harsh environments [318]. Integrating these approaches may offer a sustainable and efficient solution for mitigating antibiotic pollution and resistance.

Photocatalysis, an advanced oxidation process (AOP), has demonstrated high efficiency in degrading antibiotics and ARB by generating reactive oxygen species (ROS) [27, 141]. In addition, enzyme-based biocatalytic systems offer a promising approach for breaking down persistent antibiotic compounds [40, 156]. Microalgae-mediated treatments have also emerged as an eco-friendly solution, utilizing bioadsorption, degradation, and accumulation mechanisms to remove pharmaceutical pollutants while enabling resource recovery [58]. Despite these advancements, challenges remain in fully eliminating antibiotics from the environment and preventing their long-term ecological impacts. Future research should focus on optimizing integrated remediation technologies and assessing their large-scale applicability to ensure sustainable pollutant removal [59, 163]. Further research is needed to optimize microbial consortia for the efficient removal of antibiotics and pharmaceutical pollutants while minimizing the risks associated with ARG proliferation.

Challenges in field application of bacterial consortia

Comprehending and manipulating microbial relationships within manufactured, artificial, or semi-synthetic consortiums is intricate, requiring an understanding of compatibility between species and potential synergistic and antagonistic interactions [256]. The evolving regulatory framework for microbial consortia items in agriculture may pose challenges in obtaining commercial approvals due to uncertainties and safety concerns [10]. Enhancing the generation of microbial consortiums for large-scale applications Agriculture has logistical and economic issues, necessitating uniformity in formulas and cost-efficient production to enhance the worldwide supply chain and facilitate technological transfer [79].

In addition, the continued viability and effectiveness of microbial consortia are essential across various field settings, as environmental influences, competition with Indigenous microorganisms, and other unknown variables might influence their effectiveness over time [203, 210]. Even if microbial consortia demonstrate high pollutant degradation efficiency in laboratory conditions, their performance may decline in real-world applications due to environmental stressors such as pH fluctuations, temperature variations, and nutrient limitations [33, 168]. Moreover, synthetic consortia may struggle to establish themselves in native microbial ecosystems, where competition with indigenous microbes can limit their survival and efficiency [168, 222].

Furthermore, the introduction of synthetic and semi-synthetic microorganisms into the surroundings prompts ethical and cultural apprehensions, necessitating a balance between potential benefits and unintended consequences while also addressing public perception [43]. One of the major concerns is the risk of horizontal gene transfer (HGT), where engineered or synthetic genes from introduced bacteria could transfer to native microbial populations. This could lead to unforeseen ecological consequences, such as the spread of antibiotic resistance genes or the disruption of natural microbial processes [64].

Therefore, microbial consortia implementation necessitates interdisciplinary collaboration among microbiologists, geneticists, agronomists, and regulatory bodies to bridge gaps and ensure effective development and application [243]. Stringent biosafety measures and containment strategies, such as the development of "kill-switch" mechanisms in genetically modified bacteria, are being explored to prevent unintended environmental persistence [275]. However, challenges remain in ensuring these containment systems function reliably under diverse environmental conditions.

Bioremediation utilizes microorganisms to degrade or transform contaminants, offering a sustainable and cost-effective alternative to traditional chemical and physical remediation methods. While bioremediation minimizes environmental disruption and aligns with eco-friendly practices, its effectiveness depends on environmental conditions, and it can be slower than chemical treatments. In contrast, chemical and physical methods provide rapid contaminant removal and broad applicability but are often expensive, environmentally intrusive, and may generate secondary pollution. Comparative studies highlight the strengths and limitations of both approaches, emphasizing that integrating bioremediation with chemical or physical treatments can enhance overall efficiency [22, 30, 31, 52]. Table 3 provides a comparative overview of these remediation methods, summarizing their advantages, limitations, and suitability for different contamination scenarios.

Large-scale application of bacterial bioremediation faces cost barriers, regulatory complexities, and scalability challenges [252]. Engineering microbial consortia requires precise control over microbial interactions, which may be difficult in field conditions [206]. In addition, the introduction of genetically modified bacteria raises concerns about potential unintended ecological effects, such as horizontal gene transfer to native microbial populations [145]. Furthermore, concerns over long-term ecological impacts, such as the potential for synthetic bacteria to disrupt natural microbial communities or introduce new metabolic pathways into ecosystems, remain largely unexplored. Long-term monitoring and environmental impact assessments will be crucial for ensuring that microbial synthetic consortia do not cause irreversible changes to soil and water microbiomes [73, 155].

Environmental challenges for maintaining microbial activity in diverse ecosystems

Climate change and human activities are intensifying environmental change, testing the resilience of aquatic and terrestrial ecosystems, with diverse drivers affecting different ecosystems [376]. Europe has almost 650,000 terrestrial sites anticipated to be affected [277, 278], while an evaluation of 1511 marine sites in Europe indicated that dangerous compounds taint 93% [187, 255]. Air, water, and soil pollution levels are increasing, with climate change, resource overexploitation, and pollution compounding the detrimental effects on ecological purposes [102]. Consequently, there is increasing interest in comprehending the cumulative consequences of compounded disturbances in global ecosystems, as interactions among many disturbances can produce intricate and nonadditive outcomes [281].

Microorganisms are essential to ecosystem activities, especially in biogeochemical cycling, influencing marine, clean water, and land ecosystems [238]. Despite microbial diversity not declining, recent studies suggest that the loss of microbial taxa can impair ecosystem functions [298, 301, 390]. The significance of biological diversity in ecosystem functioning is increasing as multifunctionality is considered—a trend consistent across aquatic and terrestrial ecosystems [76]. Reestablishing microbial communities following perturbations is a crucial component of the many processes that change how ecosystems work in response to disturbances caused by changing climate conditions and more people doing more activities [222]. Most research on global environmental change concentrates on one or two variables, rendering the impacts of

Table 3 Comparing bioremediation with chemical and physical treatment methods

Aspect	Bacterial bioremediation	Chemical/physical treatments
Mechanism	Uses bacteria to degrade or transform pollutants naturally	Uses chemicals or physical processes to remove or neutralize contaminants
Speed of action	Slower, may take weeks to years	Rapid, often takes days to weeks
Cost	Generally lower, especially for large-scale applications	Generally higher, requires specialized equipment and labor
Environmental impact	Eco-friendly, utilizes natural processes	May introduce secondary chemicals into the environment
Effectiveness for organic pollutants	Highly effective, microbes can break down hydrocarbons and pesticides	Highly effective for synthetic and persistent pollutants
Effectiveness for heavy metals	Moderately effective, certain bacteria can transform heavy metals into less toxic forms	Highly effective, especially through precipitation, soil washing, or stabilization
Applicability	Limited by environmental conditions such as temperature, pH, and oxygen availability	Broadly applicable to various contaminants
Scalability	Scalable but requires site-specific adaptation	Scalable but may require high operational costs
Risk of secondary pollution	Minimal, as it relies on natural degradation	Potentially high, some treatments generate toxic by-products
Site disruption	Minimal site disruption as it can be conducted in-situ	Often involves excavation, leading to habitat destruction
Integration potential	Can be combined with chemical treatments for enhanced effectiveness	Can be used alongside bioremediation to degrade residual contaminants

compounded perturbations on the composition of microbial functions ambiguous and contentious [51].

The efficiency of microbial bioremediation can be hindered by various abiotic and biotic stressors, including fluctuations in pH, temperature, salinity, and oxygen availability [401, 402]. In addition, the presence of competing native microbial populations and inhibitory compounds such as heavy metal ions or xenobiotics may suppress bacterial activity [295]. These factors necessitate further research on microbial adaptation strategies and the use of bioengineering approaches to enhance bacterial resilience in contaminated environments.

Potential environmental risks of bioremediation byproducts

Bacterial bioremediation is a sustainable approach to degrading environmental pollutants. However, certain metabolic pathways can produce intermediate byproducts that may be more toxic than the original contaminants. For instance, the degradation of oil-derived products can lead to the generation of toxic byproducts, posing challenges for microbial communities involved in the cleanup process [106]. Similarly, incomplete breakdown of PAHs by *Pseudomonas* spp. can result in the accumulation of benzo[a]pyrene, a known carcinogen [305, 306, 319]. In the context of heavy metal remediation, bacteria often transform toxic metals into less harmful forms through processes such as oxidation and reduction. For example, bacteria such as *Micrococcus* sp. and *Acinetobacter* sp. can oxidize hazardous arsenite (As^{3+}) into the less toxic arsenate (As^{5+}), thereby reducing its toxicity [25]. However, certain transformation processes can inadvertently produce byproducts that may pose environmental concerns.

The microbial breakdown of persistent organic pollutants (POPs), such as polychlorinated biphenyls (PCBs), involves complex enzymatic reactions. Incomplete degradation can result in the accumulation of toxic intermediates. For instance, certain bacterial strains can degrade PCBs into less chlorinated congeners, which may still exhibit toxicity and persist in the environment. To address the potential risks associated with harmful byproducts, research is focusing on optimizing bacterial metabolic pathways to ensure complete degradation of pollutants. Advancements in genetic engineering and molecular biology have led to the development of microbial strains with enhanced degradation capabilities, reducing the formation of toxic intermediates. In addition, combining bacterial and fungal bioremediation strategies has shown promise in achieving a more comprehensive pollutant breakdown, thereby minimizing the accumulation of harmful byproducts [317, 351].

Utilizing microbial consortia, which combine multiple bacterial species with complementary metabolic pathways, can enhance the efficiency of pollutant degradation and reduce the formation of toxic intermediates. These consortia can tackle complex pollutants more effectively than single strains, leading to more complete mineralization and fewer harmful byproducts [86, 87]. Environmental conditions such as pH, temperature, and nutrient availability significantly impact microbial activity, and the pathways utilized during bioremediation.

Optimizing these factors can steer microbial metabolism toward complete degradation of pollutants, thereby minimizing the accumulation of toxic byproducts [217, 218]. Recent technological advances have improved the ability to monitor and control byproduct formation during bioremediation. Techniques such as metagenomics and proteomics allow for a deeper understanding of microbial communities and their metabolic pathways, enabling the development of strategies to mitigate the production of harmful intermediates [408, 409].

Challenges and ecological concerns in implementing bacterial consortia for industrial-scale bioremediation

Implementing bacterial consortia for bioremediation at an industrial scale presents several challenges, particularly in maintaining stable microbial interactions, adapting to environmental variability, and addressing regulatory concerns [4]. In addition, the application of synthetic biology in bioremediation raises ecological and ethical concerns related to genetic stability, horizontal gene transfer, and unintended environmental impacts [63].

One of the key challenges in utilizing bacterial consortia for bioremediation is the complexity of microbial interactions. Microbial consortia often involve multiple species working synergistically to degrade pollutants, but their interactions—including competition, cooperation, and signaling—must be carefully managed to maintain stability and efficiency [231]. The functional differentiation and metabolite exchange within these consortia enable complex metabolic tasks, yet the underlying mechanisms governing microbial interactions require further investigation for effective industrial applications [37]. Environmental variability poses another significant challenge. Industrial sites often experience fluctuating conditions such as changes in temperature, pH, and contaminant concentrations, which can impact microbial performance [4, 369, 370, 372]. To address this, robust microbial strains must be selected or engineered for resilience in dynamic environments [61].

Scaling up laboratory-based bioremediation processes to industrial applications also presents difficulties. Maintaining microbial activity, ensuring sufficient nutrient supply, and achieving effective distribution of bacterial consortia in contaminated sites are crucial factors [103]. In addition, synthetic microbial consortia have been explored to enhance biodegradation, yet their success in real-world applications remains limited by challenges in controlling their behavior outside of controlled laboratory settings [74]. Regulatory and public acceptance issues further complicate the deployment of bacterial consortia for bioremediation. The use of genetically modified organisms (GMOs) or engineered microbial consortia raises regulatory hurdles due to biosafety concerns [166]. Public concerns regarding the release of synthetic microbes into open environments necessitate transparent risk assessments and active engagement with stakeholders to ensure responsible implementation [74]. The use of (GMOs) in bioremediation presents ethical and regulatory challenges, requiring a balance between innovation and biosafety. Regulations vary globally, with the U.S., E.U., and China adopting different levels of restriction, highlighting the need for standardized frameworks [342]. Traditional risk assessments focus on immediate ecological effects, but long-term monitoring of genome stability and horizontal gene transfer remains insufficient [108]. Ethical concerns about human intervention in microbial evolution and public opposition further hinder adoption, emphasizing the need for transparent risk communication and public engagement [74].

On turn, the application of synthetic biology in bioremediation introduces ecological risks, particularly concerning the unintended spread of synthetic organisms. Engineered microbes may persist beyond their intended application sites, leading to unpredictable interactions with native microbial communities and potential ecological imbalances [198, 201, 202, 252]. Horizontal gene transfer is another major concern. Synthetic genes introduced into engineered microbes may transfer to native microorganisms, potentially resulting in the spread of novel traits with unforeseen ecological consequences. For example, the transfer of engineered metabolic pathways could alter nutrient cycles or create unintended selective pressures within microbial communities [166, 172]. The evolutionary stability of synthetic organisms also remains uncertain. Once released into the environment, these microbes may undergo mutations or genetic recombination, leading to unexpected traits that could confer survival advantages and make them invasive or harmful to existing ecosystems [4, 252]. Ethical and biosafety considerations further complicate the implementation of synthetic microbial consortia. The deliberate release of engineered microbes into natural systems raises ethical concerns regarding human intervention in microbial ecosystems. Robust containment strategies and regulatory oversight are essential to prevent unintended environmental impacts and ensure that bioremediation efforts remain sustainable and responsible [74].

Potential solutions such as encapsulation of bacteria for stability and efficacy enhancement

Over the past century, the chemical industry has rapidly progressed in compound synthesis, resulting in the inadvertent, intentional, or waste-related release of these substances into the environment, thereby creating significant ecological, economic, and public health issues [247]. Nonetheless, it is not a primary method for addressing the environmental contingencies of diverse nations, as there remains considerable scope for enhancement [284]. There are some problems with bioremediation at the in-situ level in maritime environments. These include a lack of food, competition for carbon sources with native microbes, spreading or diluting, changes in pH and temperature, and the pollutant's bioavailability [217, 218]. Adding microorganisms (bioaugmentation) to soils that need to be cleaned up on a large scale may not always be the best way to do it because it can change the geochemical content and the native microbial community over time [48].

One way to fix the problems with in-situ bioremediation is to immobilization of microorganisms. This protects both the organisms and the environment while bioaugmentation of polluted areas takes place in less-than-ideal conditions [30, 31]. However, Cai et al. [48] link certain microbial activities, such as nitrogen transformation, to pollutant removal processes. Therefore, the supplementation of nutrients, known as biostimulation, continues to be crucial for enhancing bioremediation [357]. This is due to the imbalanced C/N/P ratio that occurs after pollutant spills [228].

Entrapment involves capturing microorganisms in gel-like materials, creating a network-like environment that prevents external threats and biomass washout while maintaining microbial capacity [7]. Entrapment is a widely used technique for cell immobilization due to its mechanical strength, stability, and durability, but it often faces limitations in mass transfer [323]. Encapsulation, also known as encapsulation or microencapsulation, is a cell immobilization technique that covers, coats, or traps microorganisms using a semi-permeable protective film, facilitating better nutrient and substrate transfer (Wang et al. 2020. The encapsulation process consists of three essential steps: The initial phase involves the creation of an encapsulating solution containing a minimum of one active component (encapsulating substance or carrier. In the second stage, the encapsulating solution combines with the microorganisms or designated material for encapsulation,concurrently, the encapsulation process generates capsules in the third stage [294]. Two essential factors must be considered in microencapsulation that affects the properties of the capsules: the encapsulation technique and the encapsulating substance. Furthermore, these two elements must be congruent with one another and with the encapsulated microorganisms [111].

Advances in genetic engineering for improved bioremediation

Bioremediation is a crucial intervention method utilizing living organisms—primarily bacteria and fungi—to reduce soil and water contaminants [261]. Eco-friendly strategies involve encouraging microorganisms to transform contaminants into sustainable energy, producing harmless byproducts such as clean water and carbon dioxide [192]. The sustainability of bioremediation depends on optimal temperature, nutrients, and microbial food sources, as these factors can slow remediation and contamination cleanup [30, 31]. Bioremediation strategies include biostimulation, bioaugmentation, and soil catabolism. Biostimulation stimulates microbes to grow, while bioaugmentation adds bacteria on-site for faster remediation. Extrinsic bioremediation uses native microbiomes to convert toxic materials into non-toxic products. Implementation can occur in-situ or ex-situ, depending on climate and soil density. Ex-situ bioremediation may require excavation and cleaning, incurring additional costs [139].

Bioremediation offers numerous benefits over other cleanup methods, including avoiding human interference and minimizing disturbance to nearby communities [15]. Microbial processes are crucial in decomposing organic pollutants, utilizing various catalytic mechanisms to break down complex chemicals into simpler, safer components [99]. Microbial removal involves sequential cleavage of organic pollutants, converting them into inoffensive compounds, and forming the basis of bioremediation systems, a safe alternative to traditional techniques for ecosystems [166]. The study shows that engineered microbial communities could be used to clean up organic pollutants. It shows how synthetic biology and genetic engineering can be used to improve the removal abilities of these strains [99]. Scientists can modify microorganisms' genetic backbones to enhance pollutant removal efficiency, promote enzyme production, and provide environmental resistance, making them powerful tools for remediation campaigns [265]. Cytochrome P450 monooxygenases, dehydrogenases, and hydrolases are enzymes that change organic pollutants into intermediates. This shows how flexible engineered microbial communities can be [80]. Biodegradable compounds are the only ones for which bioremediation is effective because it requires rapid and complete contaminant removal, which may not be possible for nonbiodegradable substances [325].

Genetically modified *Pseudomonas putida* to increase efficiency in degrading complex hydrocarbons

Environmental biotechnology uses genetically modified microorganisms to clean up harmful chemicals, xenobiotics, and pesticides. This is how synthetic biology is changing environmental problems [265]. Understanding natural microbial communities is crucial for creating synthetic microbial communities and organisms [257]. Bioremediation involves creating an artificial microbiome with functionally specialized species using a synthetic microbial community [19]. Two microorganism species interact to influence this community, it serves as a model framework with fundamental, working, and ecological dimensions [313]. Cooperation is crucial for community organization and functioning in social microbial populations [396]. The synthetic community, a result of designed collaboration, determines its impact on dynamics. Environmental manipulation, such as gene removal and insertion, can manipulate cooperation [84]. Synthetic microbial communities, manufactured by synthetic biology, have been studied for their interaction patterns, which could be used in bioremediation systems such as pesticides and acid drainage [152]. Engineered interactions could enhance cellular activities and microbial capacity in polluted environments [2].

Singh et al. [331] demonstrated that bioremediation is the only technology that is safe, cost-effective, and sustainable for decontaminating polluted environments. Selecting and engineering microbial strains is time-consuming, scrutinizing genes linked to metal absorption, chelator synthesis, and survival in both natural and human environments. Recombinant DNA technology may modify any organism into preferred forms [138]. Inserting a host gene into a vector's genome enhances the bioremediation process, necessitating the use of various tools such as restriction enzymes, DNA ligases, T4 polynucleotide kinase, alkaline phosphatase, reverse transcriptase, and the host [263].

Using rhamnolipids, which are mostly made by *Pseudomonas aeruginosa*, made the breakdown of PAHs much better [245]. Wild-type *P. aeruginosa* opportunistic nature limits its industrial and environmental applications, and its fermentation and purification costs are high compared to synthetic surfactants [213]. Consequently, the heterologous expression of rhamnolipids in nonpathogenic strains like *Pseudomonas putida* has emerged as a focal point of research [255, 378]. It is the rhlAB operon from *P. aeruginosa* that makes rhamnolipids, and rhlRI genes control it [57]. *P. putida* KCTC overexpressed rhlABRI, producing 7.3 g/L of rhamnolipids, while Beuker et al. [34] achieved heterologous production in *P. putida* KT2440 using a synthetic promoter. The main goal of synthetic biology is to create artificial microbial groups that can break down things more easily [399].

Prospects for CRISPR-Cas9-based engineering to develop bacterial strains that can withstand harsh environmental conditions

Scientific advancements facilitate rational genetic engineering at both the global (genome) and local (gene) levels, enabling accurate insertion, deletion, or substitution of DNA fragments comprising one or more nucleotides within an organism's genome [339]. Transcription activators such as TALEN, CRISPR-Cas9, and newer genome-editing tools such as base editing and prime editing are among the most effective methods for modifying genes [324]. The CRISPR-Cas system represents an effective and accessible modification of gene technology. These tools allow for precise modifications that enhance bacterial resilience and pollutant degradation capabilities, enhancing bioremediation efforts. The DNA-binding module of TALEN exhibits sequence specificity to the host genome [287]. TALEN makes double-stranded breaks (DSBs) in DNA by attaching to it and leaving behind sticky ends. The Fok 1 cleavage domain then introduces a DSB at the specific host genome location [141].

A novel perspective on hybrid nucleases, including TALENs and ZFNs, has evolved to address molecular complexity [118]. Consequently, the CRISPR-Cas system can simultaneously edit many genes with remarkable precision. *Streptococcus pyogenes*, a bacterium, confers immunity to the virus. Crisper-derived RNA (crRNA) and trans-acting antisense RNA (trcRNA) are combined by guide RNA (gRNA) in the CRISPR-Cas system [316]. The Cas9 enzyme creates a DSB in the target DNA sequence using gRNA, awaiting bioremediation research [211]. Researchers use the CRISPR-Cas system in model organisms such as *Pseudomonas* or *Escherichia coli* [258]. Bioremediation can be done with non-model organisms such as *Achromobacter* sp. HZ01 and *Comamonas testosteroni* thanks to new knowledge about CRISPR tools and the production of gRNA [399]. Bacteria that live on contaminants are great for metabolism engineering and genomic editing due to their ability to grow and hide in environments that are toxic, resistant, and not biodegradable [268].

Recent advancements in genome editing have introduced more precise and versatile tools beyond CRISPR-Cas9. Base editing allows for targeted nucleotide substitutions without causing double-stranded breaks, minimizing unwanted mutations. Prime editing further refines genetic modifications by enabling highly precise insertions and deletions, making it particularly useful for optimizing bacterial strains for bioremediation in toxic environments. These next-generation genome-editing techniques have been successfully applied to bacteria such as *Shewanella oneidensis* and *Bacillus subtilis*, enhancing their ability to degrade organic pollutants and survive in metal-contaminated soils [47, 389]. In addition, synthetic biology approaches have enabled the rational design of microbial consortia with enhanced pollutant degradation capabilities. Engineered living materials (ELMs) have been developed using bacterial consortia integrated with biomaterials to enhance their environmental resilience. These systems have been applied in wastewater treatment, where engineered *Escherichia coli* and *Corynebacterium glutamicum* strains break down persistent contaminants, including microplastics and pharmaceutical residues [369, 370, 372, 391].

Advances in metabolic pathway engineering have further optimized bacteria for survival in harsh environments. By incorporating stress resistance genes from extremophiles, researchers have improved the heavy metal tolerance of *Cupriavidus metallidurans* and *Pseudomonas putida*, enabling them to thrive in highly contaminated soils and industrial waste sites. These modifications not only improve bacterial survival but also enhance their efficiency in detoxifying pollutants such as arsenic, cadmium, and petroleum hydrocarbons [242, 337]. In addition to bacteria, CRISPR-based genome editing has also been applied to microalgae for enhanced bioremediation. *Chlorella vulgaris* and *Nannochloropsis oceanica* have been genetically modified using CRISPR-Cas9 and TALENs to enhance their ability to remove pesticides, plastics, and heavy metals from contaminated environments. These engineered microalgae offer a scalable solution for large-scale bioremediation, particularly in water treatment applications [374].

Furthermore, CRISPR-based gene regulation (CRISPRi and CRISPRa) has emerged as a non-invasive method to optimize bacterial functions for bioremediation. CRISPR interference (CRISPRi) can selectively repress genes that limit pollutant degradation, while CRISPR activation (CRISPRa) can upregulate pathways involved in breaking down organic contaminants [332]. This approach has been successfully used in *Shewanella* species to improve electron transfer efficiency, enhancing pollutant breakdown in anaerobic conditions [47]. One of the most promising applications of CRISPR in environmental biotechnology is wastewater treatment. Genetically engineered microbial strains have been developed to target antibiotic-resistant bacteria and other persistent pollutants in wastewater. By modifying genes responsible for biodegradation pathways, researchers are improving the efficiency of microbial communities in breaking down pharmaceutical contaminants, heavy metals, and organic pollutants. However, concerns remain regarding gene leakage and potential off-target effects, necessitating the integration of biosafety mechanisms such as synthetic amino acid dependency and kill-switch designs [36].

Prospects and emerging research trends

Bioremediation, a process involving bacteria and archaea, has improved efficiency, cost, and social acceptability [13]. Recent research indicates that incorporating multiple living organisms in bioremediation enhances efficiency and results, promoting greater microbial diversity, and is widely used for removing organic and inorganic pollutants [334]. Microbial enzymes break down harmful hydrocarbons in bioremediation, and researchers are studying genetically modified microorganisms to eliminate petroleum and xenobiotic chemicals [333]. Here are some recent trends that have enhanced the potential of bioremediation.

Bioremediation in extreme environments using extremophiles such as *Deinococcus* and *Thermus* spp. for heavy metal and radioactive contamination

Microorganisms contribute to biological equilibrium and bioremediation, utilizing bacteria, algae, fungi, and yeast to eliminate pollutants from the surroundings [110]. Their plasticity and biological mechanisms render them optimal for cleanup [154]. Carbon is an essential nutrient for bacteria. We employed microorganisms from diverse settings for bioremediation. Examples of microorganisms include "*Achromobacter*, *Alcaligenes*, *Xanthobacter*, *Arthrobacter*, *Pseudomonas*, *Bacillus*, *Mycobacterium*, *Corynebacterium*, *Flavobacterium*, and *Nitrosomonas*" [327].

Microbial bioremediation, particularly with extremophilic microorganisms, offers a promising alternative to current physicochemical methods for treating toxic pollutants due to their ability to detoxify pollutants under harsh conditions [92]. Acidophilic microorganisms, such as *Thermoplasma*, *Ferroplasma*, and *Sulfolobus*, can keep pH levels stable even when the pH level is very low [330]. They do this by controlling the flow of protons through a cell membrane that is very impermeable [130]. A very important part of keeping pH levels stable is "proton pump proteins like H^+-ATPase, symporters, and antiporters from *Ferroplasma* type II and *Leptospirillum* group II" [126]. The F0F1-type adenosine triphosphate synthase in *Bacillus acidocaldarius*, *Thermoplasma acidophilum*, and *Leptospirillum ferriphilum* controls the flow of protons, and chaperone proteins and the cytoplasmic buffering capacity keep molecules inside cells safe when the pH is low [29]. Alkaliphilic microorganisms can resist high pH conditions through three key survival strategies. First, they increase proton motive force through a secondary acidic cell membrane, contributing to energy generation and pH balance [262]. Second, the sodium motive force promotes pH balance by extruding Na^+ ions, absorbing extracellular H^+ ions [113], and finally, producing organic acids for pH calibration [377].

Radiophilic microorganisms, possessing robust DNA repair systems and antioxidation mechanisms, can thrive in high-dose radiation environments, demonstrating their ability to adapt and survive under oxidative stress conditions [348]. RecA proteins from *Deinococcus radiodurans* R1, a common radiophilic bacterium, are needed to fix DNA that has been damaged by gamma rays [296]. Exposure to high doses of irradiation significantly elevates the expression ranges of many new proteins (PprA, PprM, PprI, and DdrABCDO) and DNA damage response regulons, which facilitate DNA repair and the reconstruction of damaged genomes [207]. Radiophilic microorganisms possess effective antioxidant enzymes such as "catalase, superoxide dismutase, and peroxidase", scavenging reactive oxygen species (30-fold higher activity) than radiation-sensitive bacteria such as *E. coli* and *S. cerevisiae* [116]. High levels of manganese inside cells, "polyphosphate granules, carotenoids, and pyrroloquinoline quinone" are some of the non-enzymatic factors that effectively remove ROSs and protect against protein damage. Orthophosphates, free amino acids, and small peptides protect biomolecules from ionizing radiation [229].

Bioaugmentation strategies using synthetic biology to create designer microbial consortia for targeted bioremediation

Bioremediation involves multidisciplinary approaches to environmental biotechnologies, including bioaugmentation, which involves adding microbial biomass to contaminated areas to improve pollutant removal efficiency, reduce time and costs, and adapt to the green environment [8, 9]. Bioaugmentation in controlled conditions remains challenging due to biodiversity, substrate competition, and climatic conditions [67]. Optimizing strategies requires further studies on bacteria's resistance, enzyme activity, and system robustness, also, the amalgamation of "genetic engineering, nanotechnology, and biological systems" [44]. New research on bioaugmentation looks at a variety of site-specific methods, including thermophilic reductive dechlorination [41], psychrophilic groundwater treatment, microbially induced calcium carbonate precipitation, hydrocarbon removal, bacterial remediation of pesticide-polluted soils, heavy metal removal mechanisms, autochthonous and allochthonous bioaugmentation, yeast bioaugmentation of wastewaters, and plant growth stimulation in augmented hydroponic systems [88, 118, 324].

Adding an autochthonous bacterial consortium to a municipal sewage sludge/straw substrate changed the structure of the microbe community in a big way. They found higher proportions of Firmicutes in the bioaugmented inoculum and detected *Pseudomonas putida* in all treated samples [303]. When wheat straw amendment and biological enhancement were combined, there was the most microbial metabolism, the least ammonia emissions, and the most noticeable improvement in the growth of cress seedlings [371].

Proteobacteria, *Firmicutes*, *Bacteroidetes*, *Actinobacteria*, and *Acidobacteria* dominate crude oil removal, with key enzymes detected in *Bacillus megaterium*, *Bacillus pumilus*, *Pseudomonas aeruginosa*, and *Stenotrophomonas maltophilia* [283]. Manganese-oxidizing *Pseudomonas* sp. QJX-1 is proposed for pharmaceutical drug removal [392]. *Rhodococcus* spp. are crucial in the removal of organic pollutants and nitrification in activated sludge. Their catabolic activity entails the catabolism of short- and long-chain alkanes, aromatic compounds, and aromatic compounds [134]. They exhibit high adaptability and hyperrecombination evolutionary strategies, with linear plasmids storing multiple biodegradative genes [411].

Combining bacterial bioremediation with other technologies such as phytoremediation and nanotechnology to enhance pollutant removal efficiency

Bioremediation, a cost-effective and environmentally friendly method for restoring oil-contaminated environments, is gaining momentum due to the incorporation of advanced technologies [129, 343]. Bacteria such as "*Bacillus*, *Acinetobacter*, *Rhodococcus*, *Burkholderia*, *Pseudomonas*, *Mycobacterium*, *Kocuria*, *Enterobacter*, *Arthrobacter*, *Marinobacter*, *Streptococcus*, *Staphylococcus*, *Alteromonas*, and *Achromobacter*" decompose hydrocarbon constituents from contaminated locations using enzymes with varying reaction pathways [104, 383]. Plant species can perform phytoremediation, and it entails the removal of oily constituents from polluted settings by assimilating hydrocarbon elements from soil or water and transforming them into simpler forms [148].

Bioremediation is a cost-effective and clean method for contaminant remediation, but its efficiency decreases if microorganisms or flora is incapable of enduring severe surroundings [119]. To efficiently remove hydrocarbon contaminants, bioremediation techniques, such as nanotechnology, require advancements [132]. Nano-molecules provide improved effectiveness and reactivity owing to their superior surface-to-volume ratio compared to bulkier countermolecules [181]. Research demonstrates that various nanomaterials can effectively remediate petroleum oil pollution [240]. Combining nanotechnology and bioremediation results in nanobioremediation, a practical solution for oil-polluted sites with attractive nanoparticle properties that minimizes limitations and improves waste removal [272, 273]. Researchers are increasingly using NPs in various applications, but their chemical synthesis can be hazardous [361]. To address this, researchers are focusing on green NP synthesis, using biological components such as biometabolites, plant parts, and microbes to create sustainable NPs for sustainable use [140].

Nanobioremediation enhances pollutant degradation by improving microbial interactions with nanomaterials, though their effects can be both beneficial and toxic. Proper NM selection maximizes efficiency while minimizing risks, contributing to environmental sustainability [360]. Nanotechnology improves bioremediation by enhancing microbial activity with eco-friendly nanomaterials, reducing costs and increasing sustainability. Enzyme-nanotechnology integration has further enhanced enzyme activity and reusability, outperforming conventional methods [224]. Nanoparticles improve pollutant solubility and bioavailability, enhancing microbial degradation. AI-driven nanobioremediation is emerging as a tool to optimize site rehabilitation [272, 273]. Biosynthesized nanoparticles provide a cost-efficient, sustainable solution by enhancing microbial activity and reducing energy consumption [267, 269]. Nanomaterials improve pollutant removal rates due to their high surface area and sequestration properties. Materials such as nanoscale zero-valent iron (nZVI), carbon nanotubes, and nano-fibers effectively remediate chlorinated compounds, hydrocarbons, and heavy metals [35]. Heavy metal contamination requires sustainable solutions. Biogenic nanoparticles offer a cost-effective alternative, as microorganisms produce nanoparticles with controlled properties. Optimizing interactions between metals, microbes, and nanomaterials ensures effective bioremediation [160]. Integrating phytoremediation with nanotechnology, biosurfactants, and oxidation processes improves pollutant destruction and degradation. Genetic engineering and chemical enhancers further optimize remediation [308].

Nanotechnology presents a promising avenue for enhancing bacterial bioremediation by improving pollutant bioavailability, increasing microbial activity, and providing novel tools for monitoring remediation processes. One emerging trend is the use of nanocarriers to deliver nutrients, electron donors, or signaling molecules directly to bacterial consortia in contaminated environments. This approach could improve bacterial survival and activity in harsh conditions, but challenges remain in scaling up production and ensuring eco-friendly degradation of nanocarriers [133]. The integration of nanotechnology with biosensors can provide real-time monitoring of pollutant degradation and microbial activity. However, field-deployable, cost-effective nanosensors for large-scale bioremediation applications are still under development [268].

Artificial intelligence (AI) is an increasingly significant technology across numerous domains due to its precision, rapidity, and accuracy. Artificial neural networks (ANN) and support vector machines (SVM) are two examples of advanced machine learning methods that are useful for environmental science because they give broad insights and make it easier to understand nonlinear data [320]. AI-based models can predict the relationship between where contaminants come from, where they are found, how concentrated they are, and their properties in a certain area. This helps environmentalists plan and carry out their work more effectively [101, 115].

Conclusion and future prospects

The increase in artificial substances, manufacturing, farming, and domestic applications has resulted in the discharge of organic and inorganic contaminants into the soil, with heavy metals being the most significant. Reducing soil pollution and promoting sustainable methods, such as removal, are crucial for environmental and health protection. Research indicates that microbial organizations have varying clearance capacities, rendering them appropriate for the bioremediation of soil pollutants. Mixed microbial consortia demonstrate superior substrate tolerance and enhanced pollutant adsorption or removal efficacy relative to a singular strain. This method employs essential intermediate chemicals to digest pollutants synergistically, generate highly efficient biosurfactants, self-regulate, stimulate bacterial proliferation, synthesize critical degrading enzymes, and augment bacterial activity. It also improves the efficiency of soil pollutant removal by enabling direct interspecific electron transfer (DIET) among dietary partnerships. This helps microbial consortiums grow and boosts bacterial activity. Microbial consortiums are very good at absorbing, breaking down, and turning soil pollutants into carbonate precipitation.

Synthetic biology offers transformative potential for bacterial bioremediation by engineering microbial consortia with optimized metabolic pathways, stress resilience, and coordinated pollutant degradation. However, a major research gap lies in designing multi-species consortia where each species performs a specialized function, leading to more efficient pollutant degradation. Advances in metabolic modeling and CRISPR-based genome editing could allow for the precise construction of consortia with complementary metabolic pathways.

Synthetic communication systems mimicking quorum sensing are needed to enable microbes to dynamically adjust metabolism based on environmental cues. In addition, robust containment strategies, such as self-destruct gene circuits, must be developed to prevent uncontrolled proliferation of engineered bacteria.

Contamination is a major global environmental issue, and while bioremediation is a cost-effective and clean treatment method, it faces challenges that need to be addressed for improved efficiency. Nanotechnology's research and advantages have been due to its extensive utilization spanning domains such as pharmaceutical shipment, agriculture, optics, aerospace, and the ecological remediation of hazardous contaminants. We extensively employ nanoparticles (NPs) to remediate polluted areas and rehabilitate ecosystems by sequestering hydrophobic contaminants, allowing efficient removal by microbes. The NBR process, combining nanotechnology and bioremediation, is attracting interest due to its superior efficiency and eco-friendly characteristics, enhancing the overall removal process. There is limited research on the synergy mechanism between NPs and biological entities in NBR processes, although the results are promising. To optimize and upgrade the process's efficiency and practical applicability, further studies on NBR in oil spill remediation are necessary.

Microbial adaptability is crucial for sustained bioremediation. Future research should explore bioencapsulation techniques, adaptive genetic engineering, and bioaugmentation strategies to overcome the limitations imposed by stress factors in contaminated environments. In addition, future research should focus on developing controlled-release mechanisms, conducting long-term ecological impact assessments, and establishing regulatory frameworks that balance innovation with environmental safety. Using bacterial consortia for bioremediation presents a promising approach to tackle environmental pollution stemming from heavy metals, hydrocarbons, and persistent pollutants. By leveraging the synergistic interactions among diverse microbial species, it is possible to enhance removal efficiency and adaptability to various environmental conditions. Continued research and development in this field will be essential for overcoming existing challenges and improving bioremediation practices.

Acknowledgements
The author thanks the Department of Biological Sciences, Faculty of Science, King Abdulaziz University, for their support.

Author contributions
The author wrote and revised the review article.

Funding
This work received no funds.

Data availability
No datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent to publications
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 22 December 2024 Accepted: 7 April 2025
Published online: 04 June 2025

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